Transform Method for Heavy-Traffic Analysis

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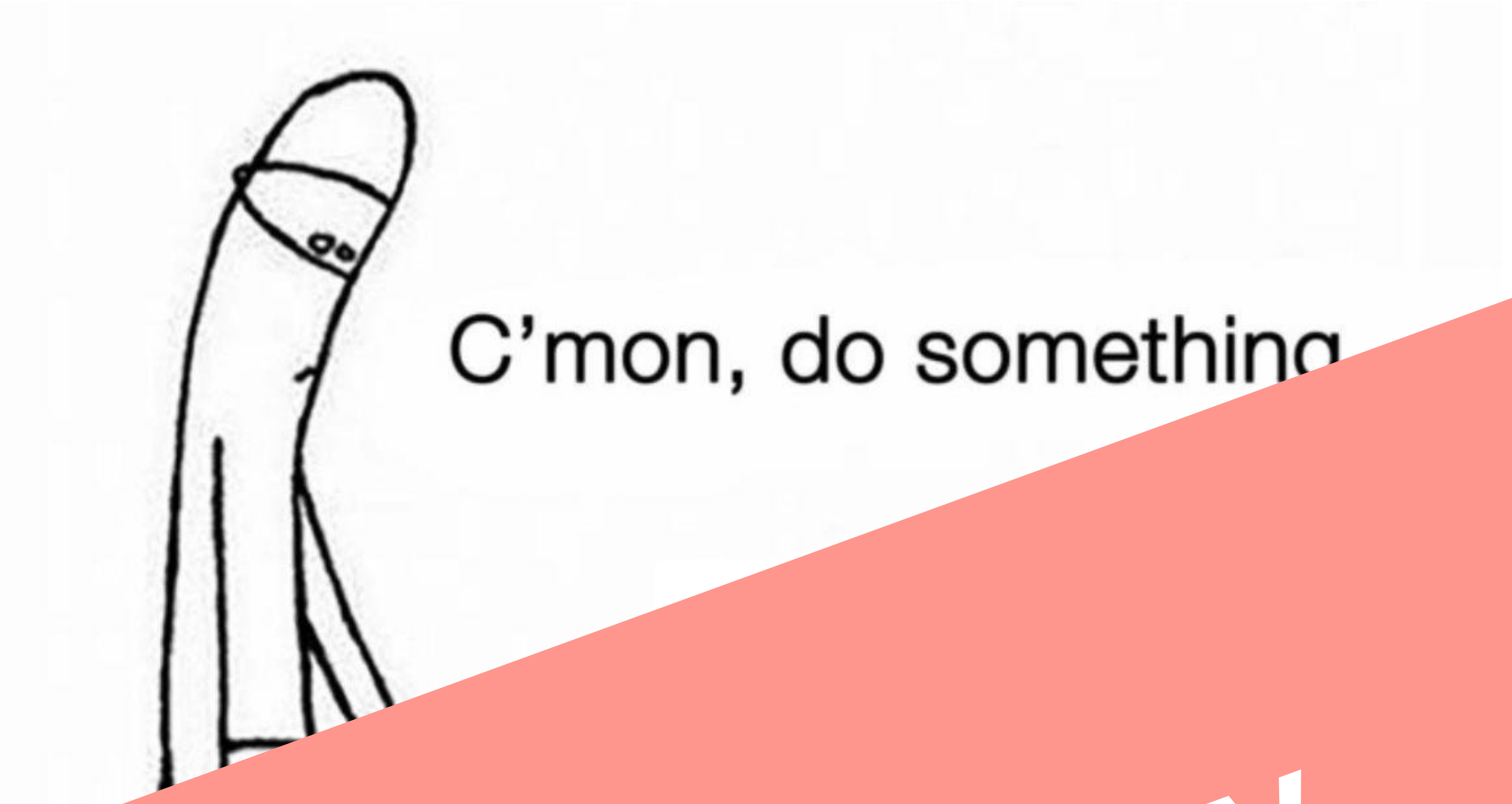
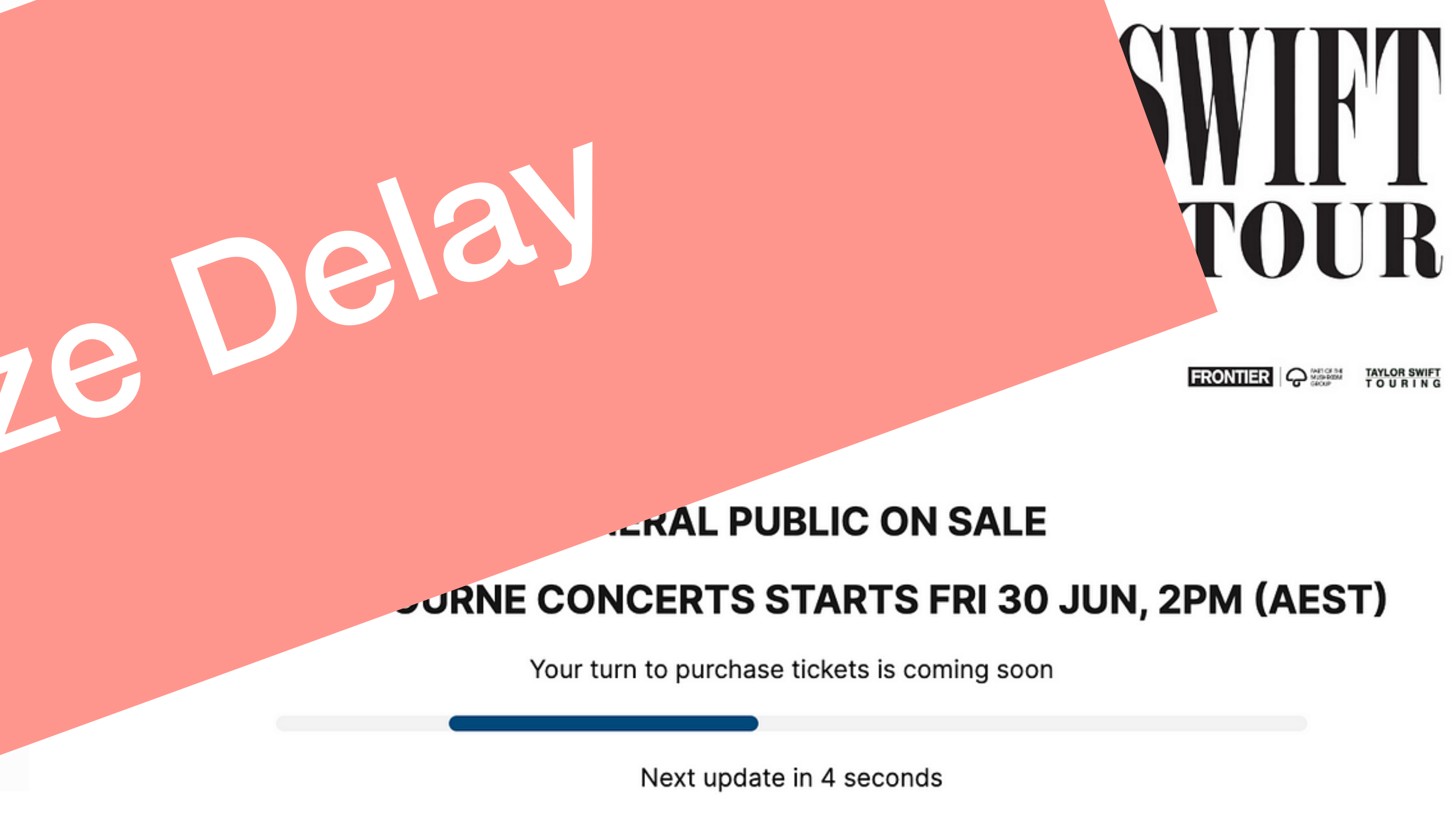
Waiting in Line



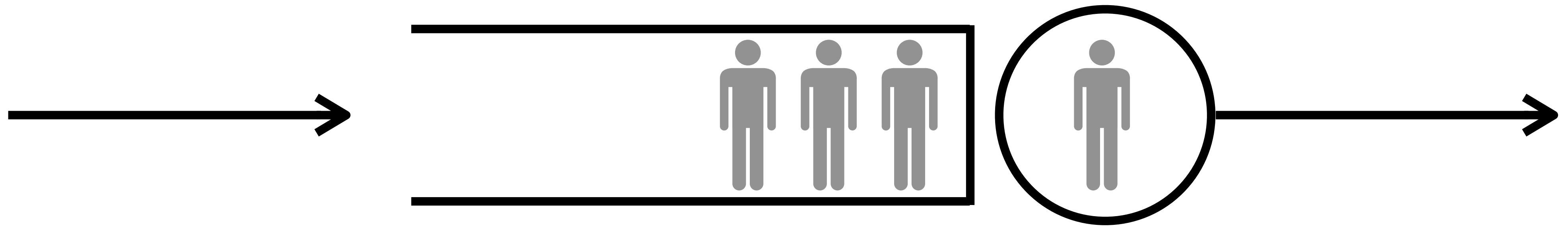


Ultimate goal: Minimize Delay

Source: TaskRabbit website

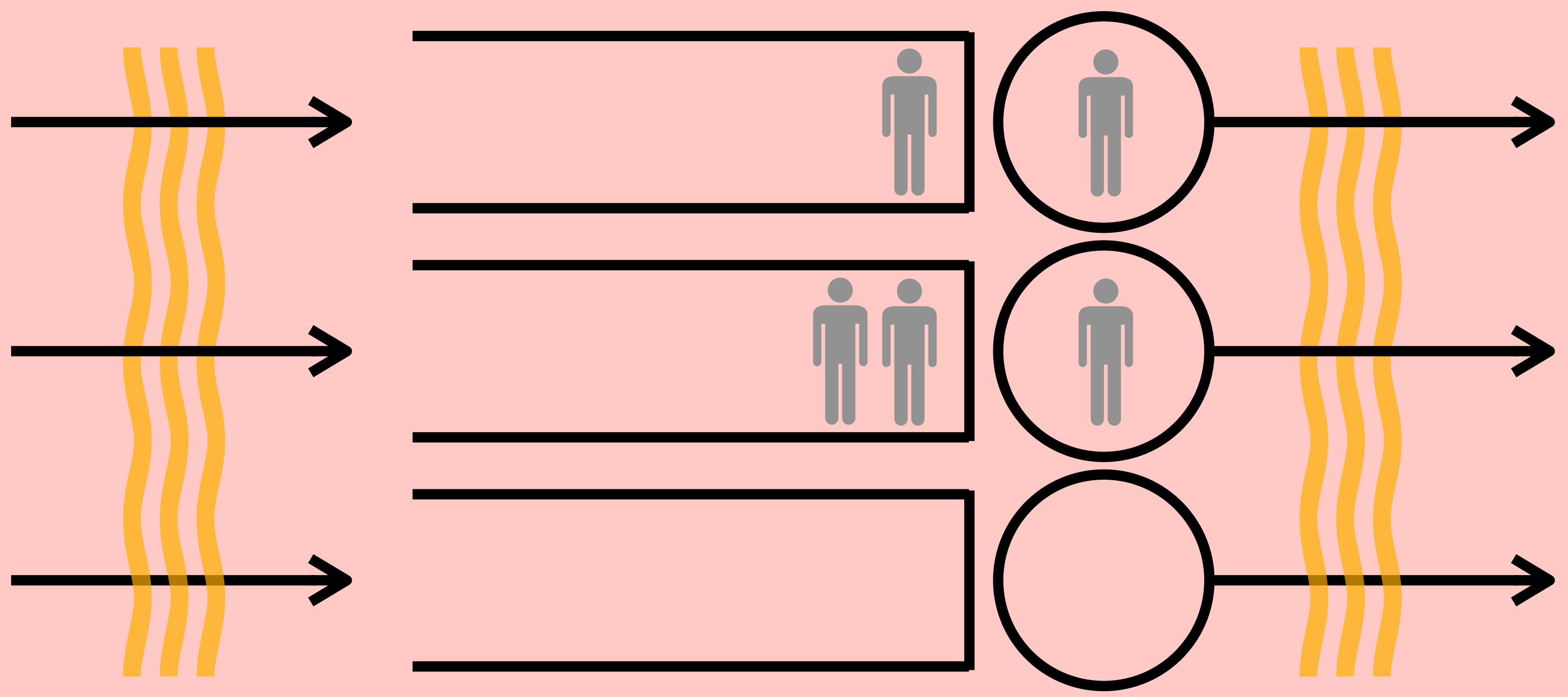
Understanding Delay

Single-Server Queue



✓ Well understood
Distribution of q is known

Multiple queues with correlated arrivals and services



This talk

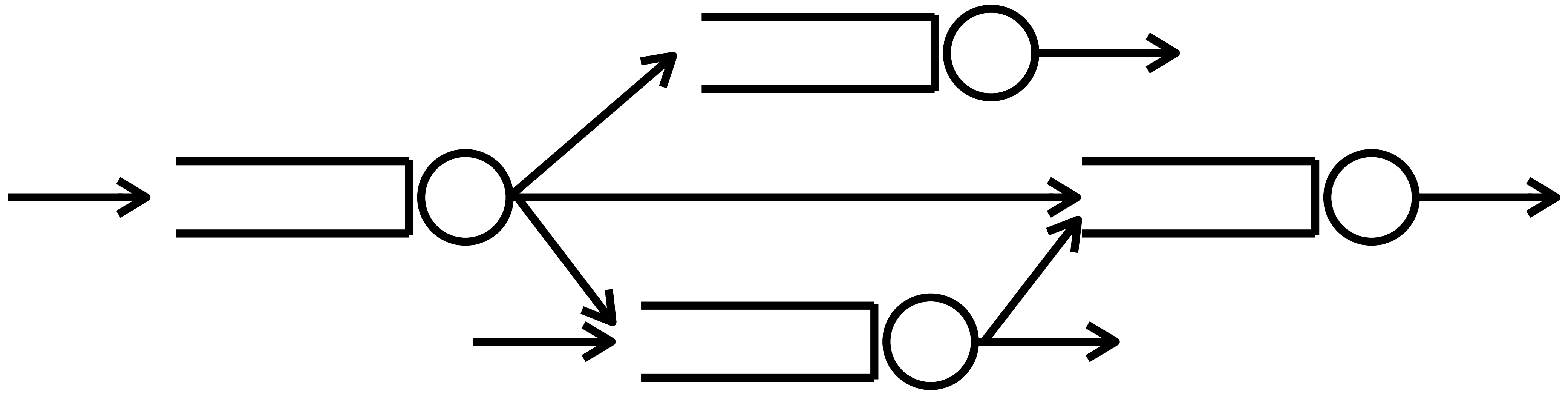
Distribution of $\boldsymbol{q}$?

Mean: $E[\boldsymbol{q}]$? Variance: $Var[\boldsymbol{q}]$?

How about $\sum q_i$?

Tail prob: $P(\sum q_i > x)$?

Networks of queues



✓ Jackson networks:
Distribution is product form

Outline

Delay Distribution and Tail Probabilities

- Asymptotic regimes in queueing theory
- Introduction to Transform Method
- Analysis of single server queue
- Systems with a single bottleneck
- Tight nonasymptotic bounds
- More general arrival and service processes

Expected Delay

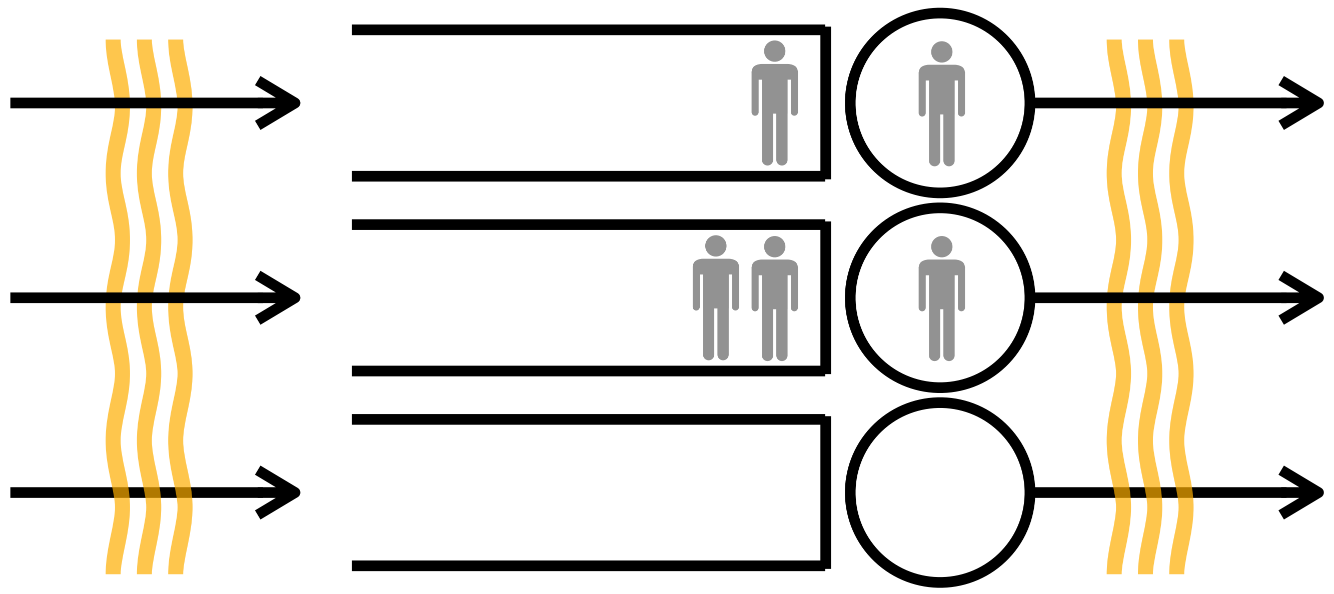
- Systems with multiple bottlenecks
- Computing the mean

Future Work

- Learning and Queueing
- Transient Analysis

In the literature…

Asymptotic Regimes



Large Deviations:
(a.k.a. rare events)
$\lim_{x\to\infty} P(wait > x)$

Ganesh (2004). Big Queues.
Glynn and Whitt (1994)
Puhalskii (1995)
Duffield and O'Connell (1995)
Dupuis and Ellis (1995)
…

Mean Field:
(a.k.a. many servers)
servers $\rightarrow \infty$

Ying (2017)
Mukkerjee et.al. (2018)
Mitzenmacher (1996, 2001)
Stolyar (2017)
…

Many-Server Heavy-Traffic:
servers $\rightarrow \infty$
& Arrival rate $\approx$ Service rate

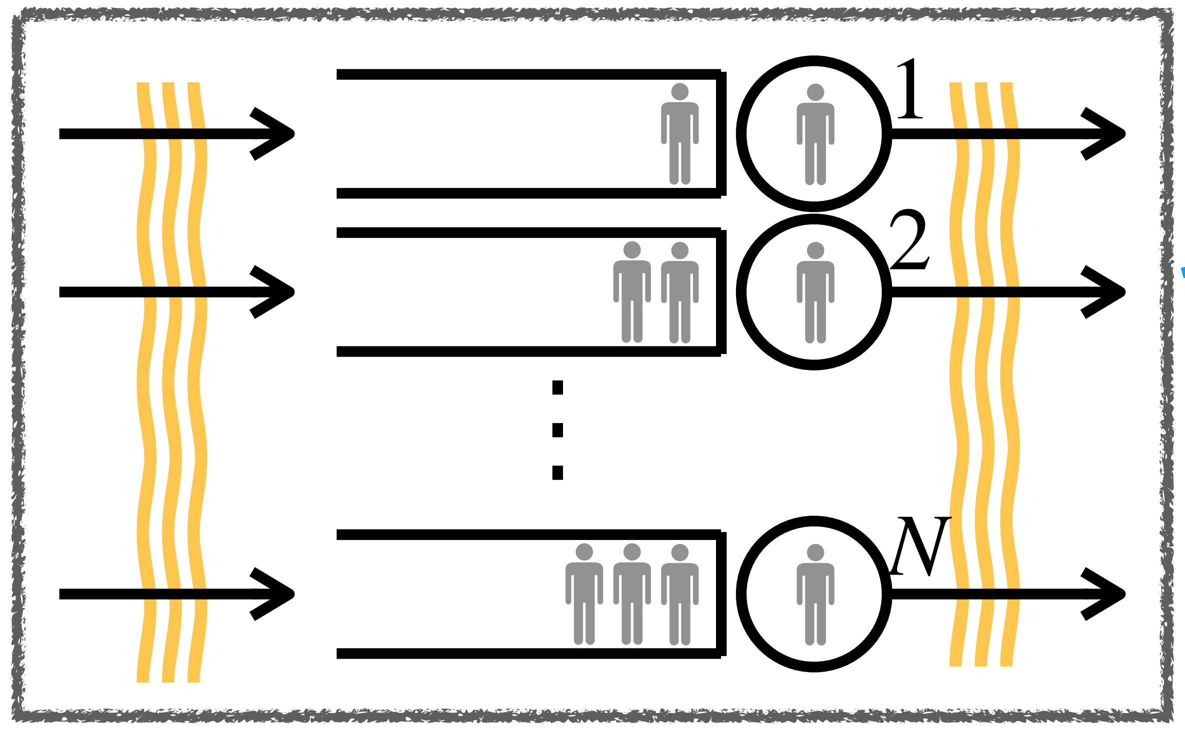
Banerjee et.al. (2019, 2020)
Liu and Ying (2019)
Braverman (2020)
Varma et.al. (2022)
HL, Maguluri (2022)
Jhunjhunwala, **HL**, Maguluri (2023)
…

Heavy Traffic:
Load system to max capacity
$\iff$ Arrival rate $\approx$ Service rate

Kingman (1962)
Harrison (1998)
Williams (1998, 2000)
Harrison and López (1999)
Stolyar (2004)
Gamarnik and Zeevi (2006)
Eryilmaz and Srikant (2012)
HL, Maguluri (2020, 2021)
HL et.al. (2022)
…

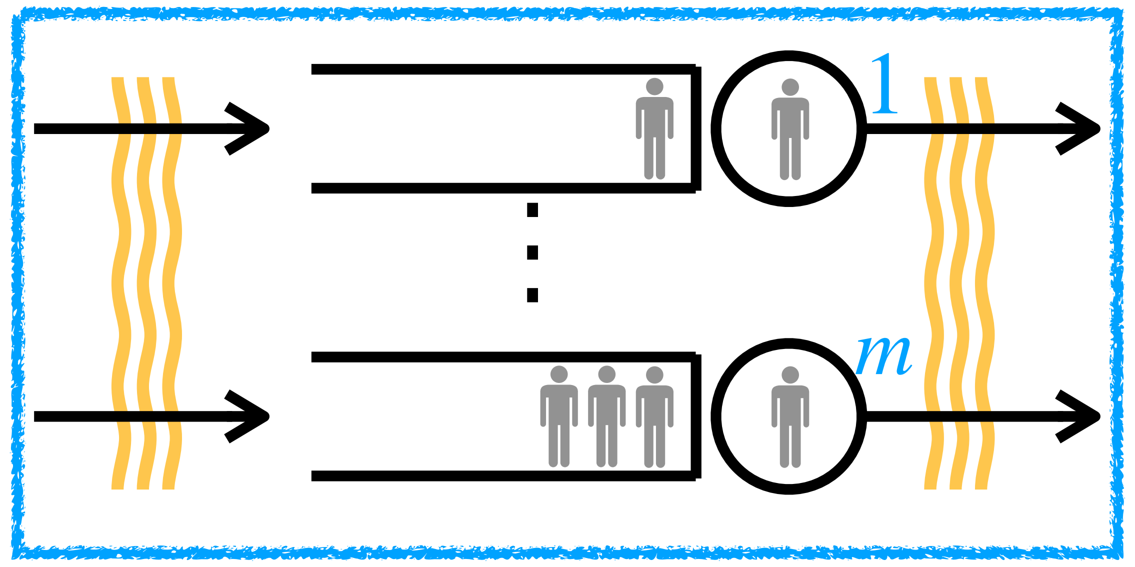
Why Heavy-Traffic Analysis?

- Arrival rate $\approx$ Service rate
- **State Space Collapse (SSC):**



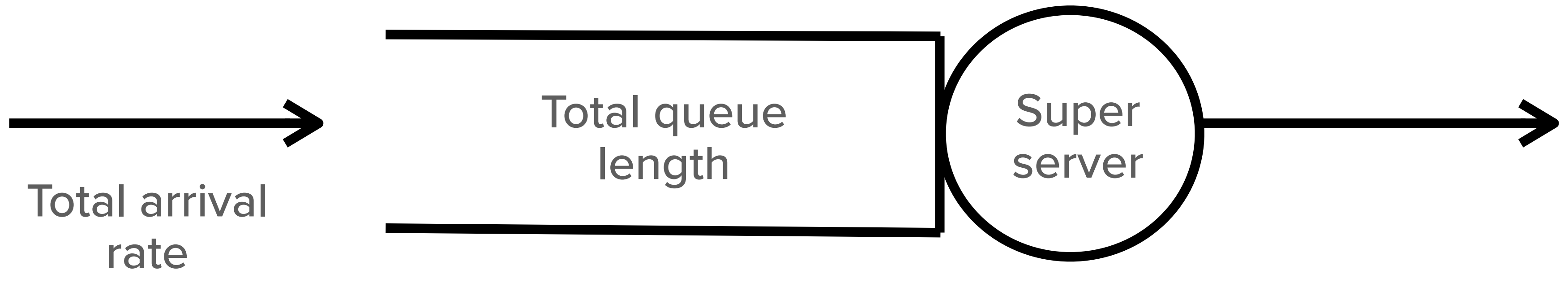
Heavy-traffic limit

$m \leq N$



Complete Resource Pooling (CRP):

- SSC to a one-dimensional subspace
- System behaves as a single server queue
- All servers “pool” together and behave as a super server



Heavy-Traffic Analysis in the Literature

Diffusion Limits Approach

- Most popular
- Introduced by Kingman (1962)
- Show convergence in distribution of queue length scaled heavy-traffic parameter ϵ

$$
\begin{tikzcd}
\epsilon q^{(\epsilon)}(t) \arrow[r, "t \to \infty"] \arrow[d, "\epsilon \downarrow 0"'] & \epsilon q^{(\epsilon)}(\infty) \arrow[d, "\epsilon \downarrow 0"'] \\
\tilde{q}(t) \arrow[r, "t \to \infty"] & q^*
\end{tikzcd}
$$

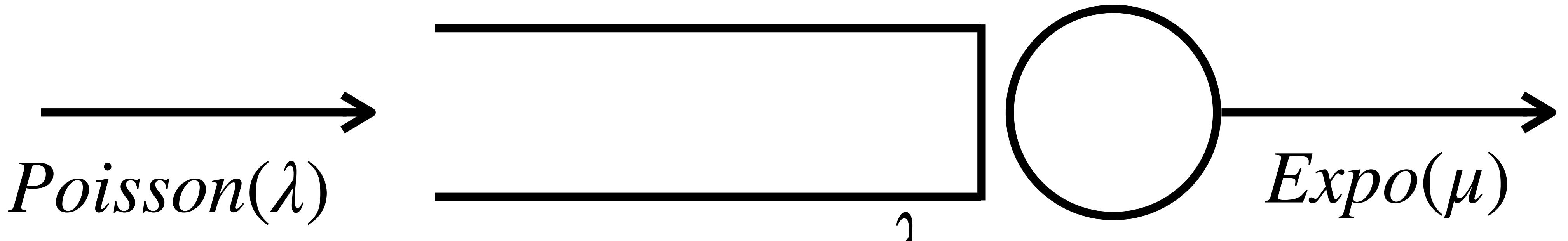
Interchange of limits

RBM

Direct Methods

- **Stein's method**
  Bound Wasserstein distance between $\epsilon q^{(\epsilon)}(\infty)$ and q^*
- **Drift Method**
  Inductively compute $\mathbb{E}\left[q^k\right]$ for $k \in \mathbb{N}$
- **BAR approach**
  Analyze $e^{\epsilon q^{(\epsilon)}(\infty)}$ and bound jumps appropriately
- **Transform Methods**
  Analyze general equation for $e^{\epsilon q^{(\epsilon)}(\infty)}$ and directly compute $\mathbb{E}\left[e^{\epsilon q^{(\epsilon)}(\infty)}\right]$
  - ✓ Tractable analysis
  - ✓ Compute distribution
  - ✓ Obtain tail bounds

Transform Method for $M/M/1$ queue



Load: $\rho = \dfrac{\lambda}{\mu}$

Drift: $\Delta V_\theta(q) = \mathbb{E}\left[e^{\theta q(t+)} \,\middle|\, q(t) = q\right]$

Step 1: Drift of exponential test function

$V_\theta(q) = e^{\theta q}, \theta \in \mathbb{R}$

Arrival　　Departure

$$\Delta V_\theta(q) = \lambda(e^{\theta(q+1)} - e^{\theta q}) + \mu 1_{\{q>0\}}(e^{\theta(q-1)} - e^{\theta q})$$

$$= e^{\theta q}\left(e^{-\theta} - 1\right)\left(\mu - \lambda e^{\theta}\right) - \mu\left(e^{-\theta} - 1\right) \underbrace{e^{\theta q} 1_{\{q=0\}}}_{= 1_{\{q=0\}}}$$

Step 2: Set drift to zero

$$\mathbb{E}\left[e^{\theta q}\right] = \frac{\mathbb{P}[q=0]}{1 - \rho e^{\theta}}$$

Step 3: Set $\theta = 0$

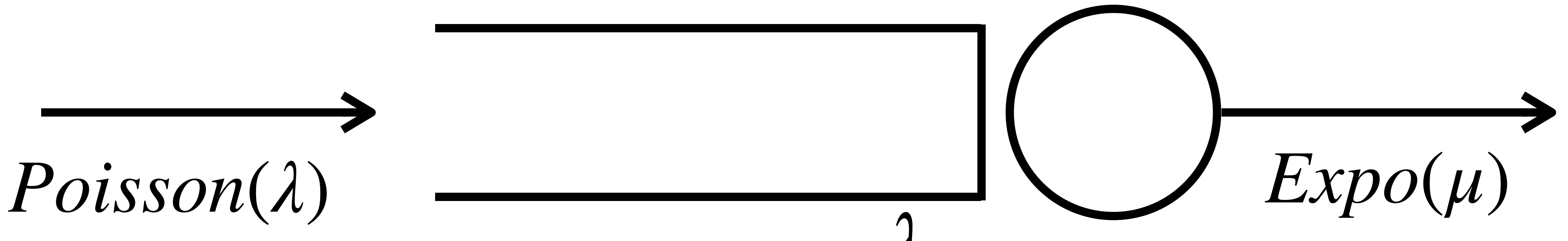
$$\mathbb{P}[q=0] = 1 - \rho$$

$$\mathbb{E}\left[e^{\theta q}\right] = \frac{1-\rho}{1 - \rho e^{\theta}}$$

$\Longleftrightarrow$ $q \sim$ Geometric$(1-\rho)$

Transform Method for $M/M/1$ queue

Why does it work?



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$$\Delta V_\theta(q) = e^{\theta q}\left(e^{-\theta} - 1\right)\left(\mu - \lambda e^{\theta}\right) - \mu\left(e^{-\theta} - 1\right) 1_{\{q=0\}}$$

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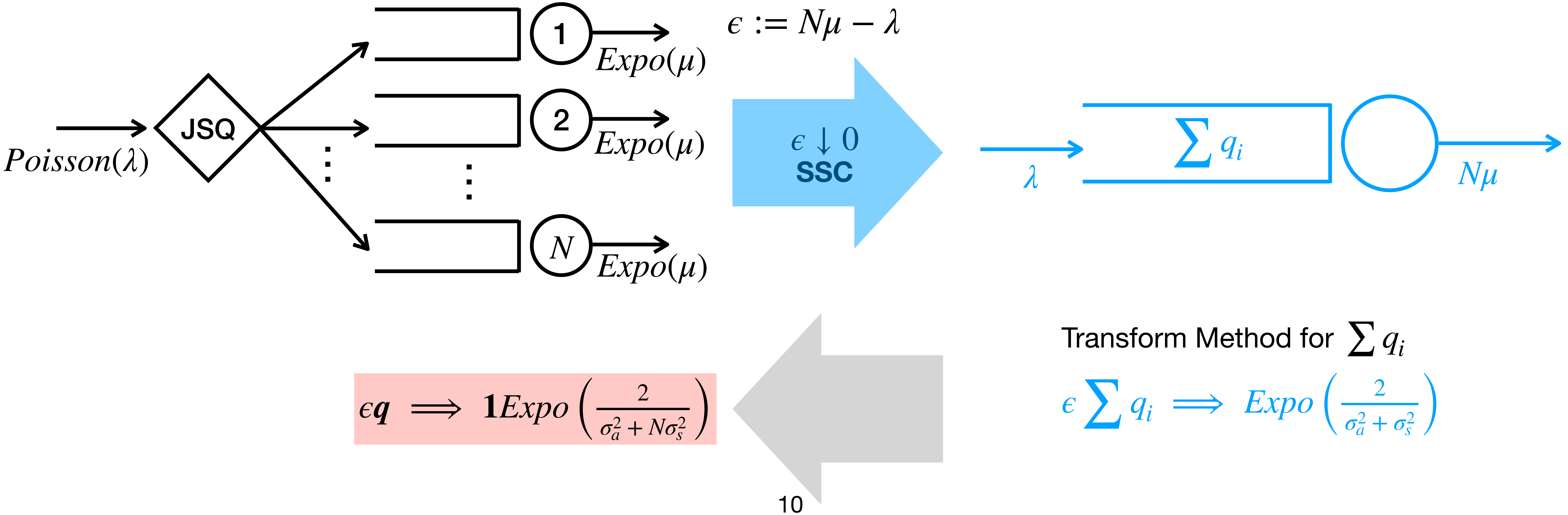
Step 3: Set $\theta = 0$

$$\mathbb{P}[q=0] = 1 - \rho$$

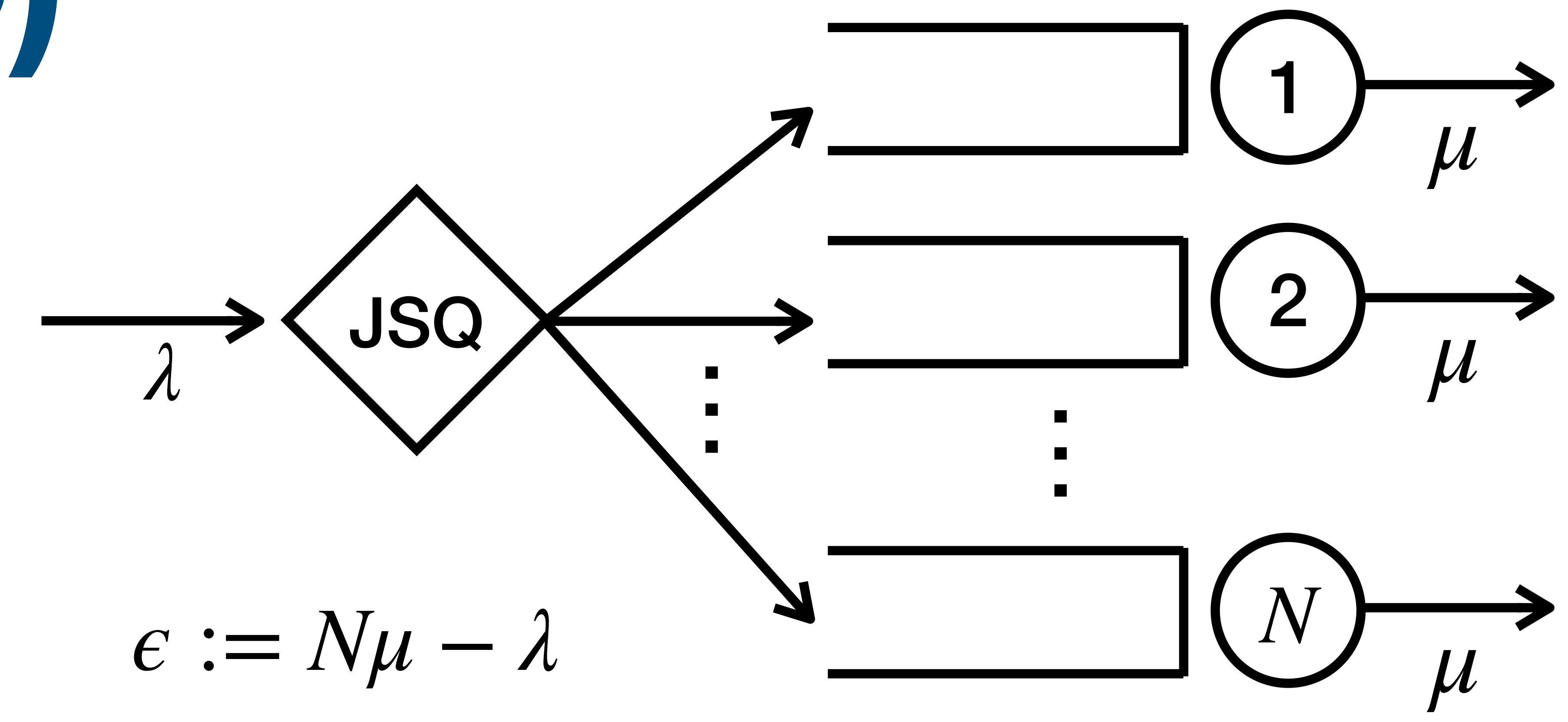
- ✓ Markov structure: $q(t+)$ only depends on $q(t)$
- ✓ $V_\theta(q) = e^{\theta q}$ yields the Laplace transform of the queue lengths
- ✓ $\Delta V_\theta(q)$'s terms only have 1 random variable
  ➡ Independent r.v. in general
- ✓ **Careful choice of test function leads to easy analysis**

Reach of Transform Method (so far)

- General arrival and service distribution: $\epsilon q \implies Expo\left(\frac{2}{\sigma_a^2 + \sigma_s^2}\right)$
- CRP systems:
  - Join the Shortest Queue (JSQ):



State Space Collapse (SSC)



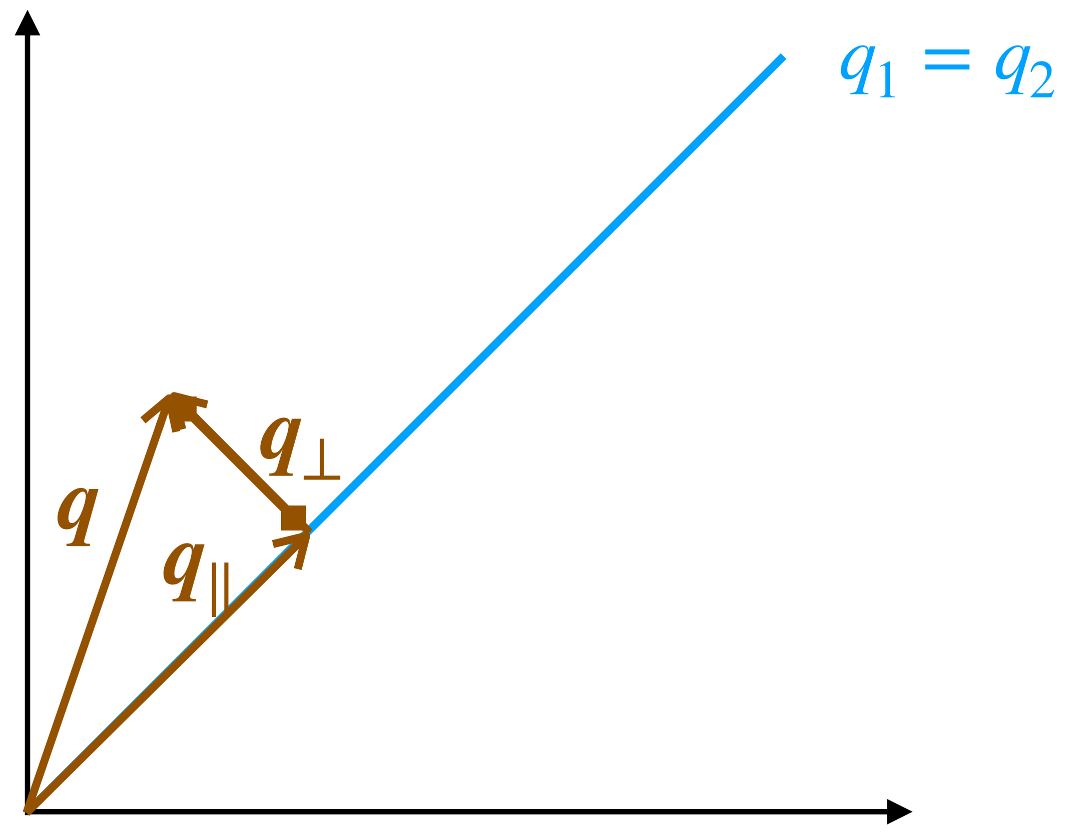
Proposition (SSC): [HL, Maguluri '20]

$\boldsymbol{q}_{\|}(k) =$ Projection of $\boldsymbol{q}$ on $\mathbf{1}$

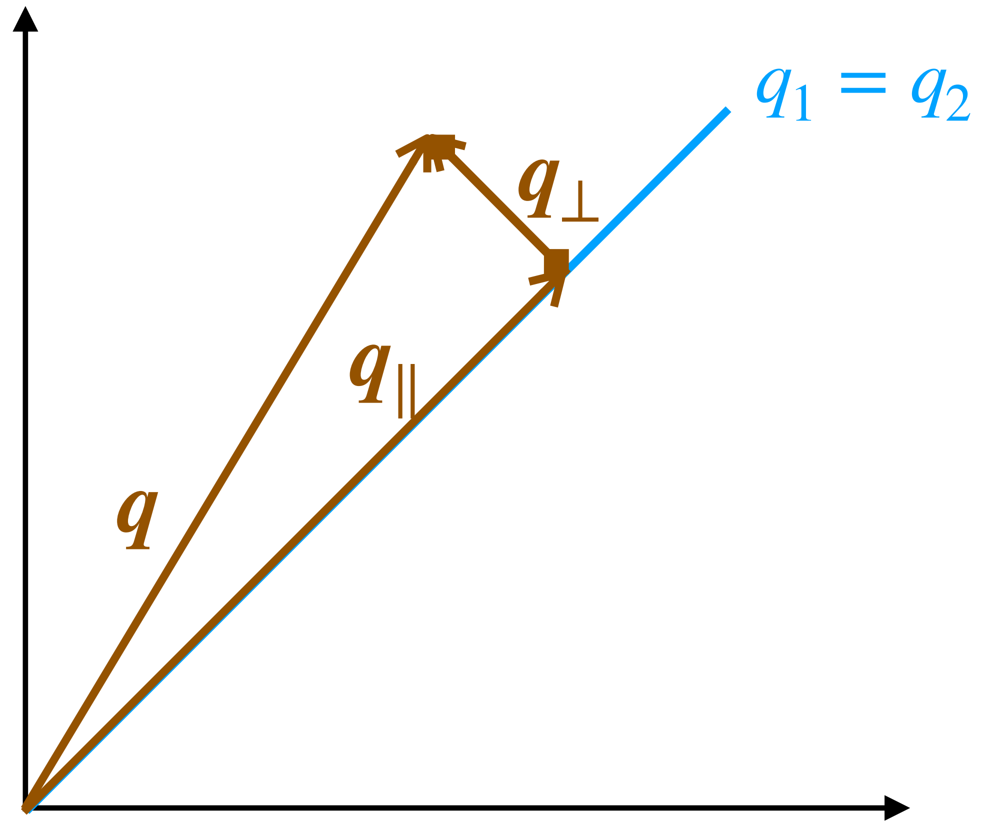
$\boldsymbol{q}_{\perp}(k) = \boldsymbol{q}(k) - \boldsymbol{q}_{\|}(k)$ error of approximating $\boldsymbol{q} \approx \boldsymbol{q}_{\|}$

Then, there exists constants M_k independent of ϵ such that for all $k = 1,2,3\ldots$

$$\mathbb{E}\left[\|\boldsymbol{q}_{\perp}\|^k\right] \leq M_k$$

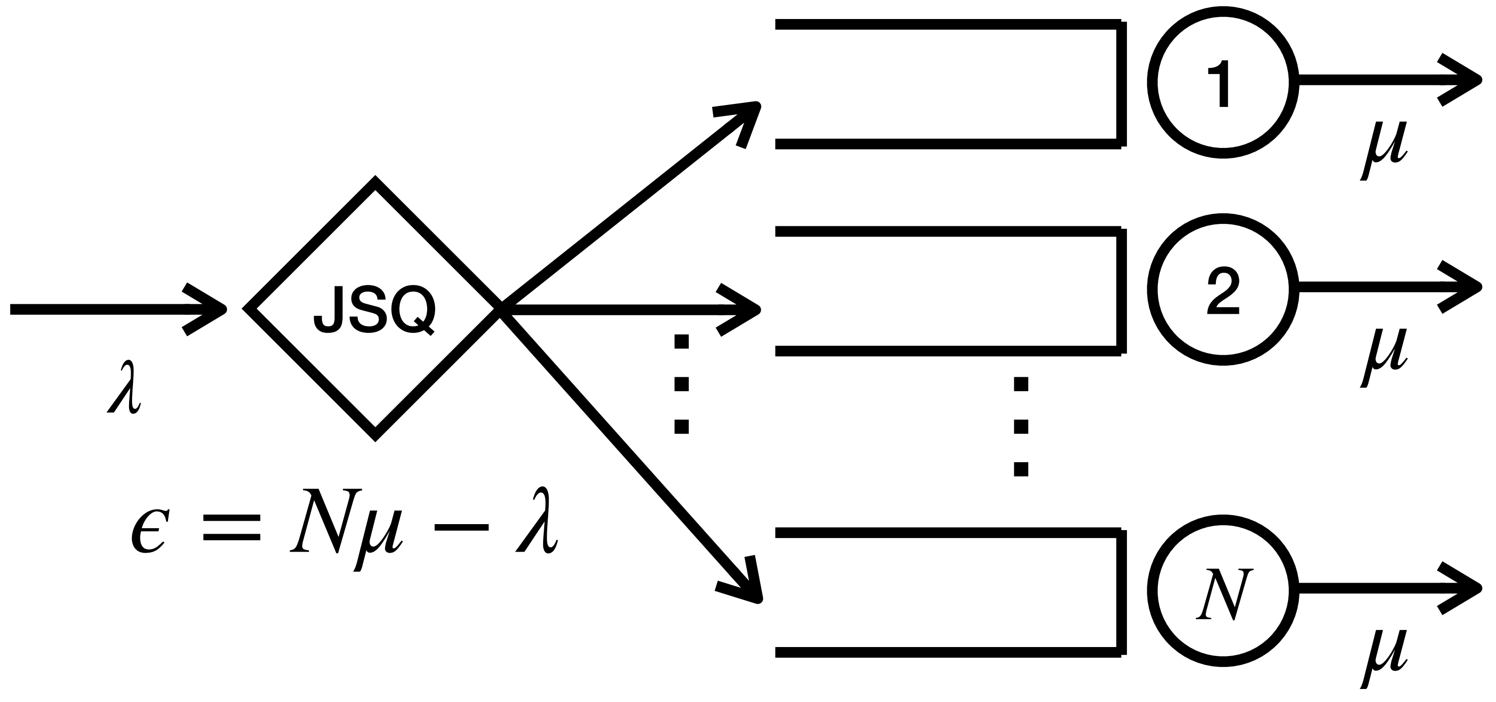


$\boldsymbol{q} \approx \boldsymbol{q}_{\|}$



Transform Method

$$\epsilon = \mu - \lambda$$



$$\epsilon = N\mu - \lambda$$

JSQ system ≈ 

Step 1: Drift of exponential test function

$$V_\theta(q) = e^{\theta q},\ \theta \in \mathbb{R}$$

Arrival — Departure

$$\Delta V_\theta(q) = \lambda\left(e^{\theta(q+1)} - e^{\theta q}\right) + \mu 1_{\{q>0\}}\left(e^{\theta(q-1)} - e^{\theta q}\right)$$

$$= e^{\theta q}\left(e^{-\theta} - 1\right)\left(\mu - \lambda e^{\theta}\right) - \mu\left(e^{-\theta} - 1\right) e^{\theta q} 1_{\{q=0\}}$$

$$= 1_{\{q=0\}}$$

Step 2: Set drift to zero

$$\mathbb{E}\left[e^{\theta q}\right] = \frac{\mathbb{P}[q=0]}{1 - \left(1 - \frac{\epsilon}{\mu}\right) e^{\theta}}$$

Step 3: Set $\theta = 0$

$$\mathbb{P}[q=0] = \frac{\epsilon}{\mu}$$

$$\mathbb{E}\left[e^{\theta q}\right] = \frac{\epsilon}{1 - (1 - \epsilon/\mu)\, e^{\theta}} \iff q \sim \text{Geometric}\left(\frac{\epsilon}{\mu}\right)$$

Step 1: Drift of exponential test function

$$V_\theta(\boldsymbol{q}) = e^{\theta \sum q_i},\ \theta \in \mathbb{R}$$

Arrival — Departure

$$\Delta V_\theta(\boldsymbol{q}) = \lambda\left(e^{\theta(\sum q_i + 1)} - e^{\theta \sum q_i}\right) + \sum_j \mu 1_{\{q_j>0\}}\left(e^{\theta(\sum q_i - 1)} - e^{\theta \sum q_i}\right)$$

$$= e^{\theta \sum q_i}\left(e^{-\theta} - 1\right)\left(n\mu - \lambda e^{\theta}\right) - \mu\left(e^{-\theta} - 1\right) e^{\theta \sum q_i} \sum_j 1_{\{q_j=0\}}$$

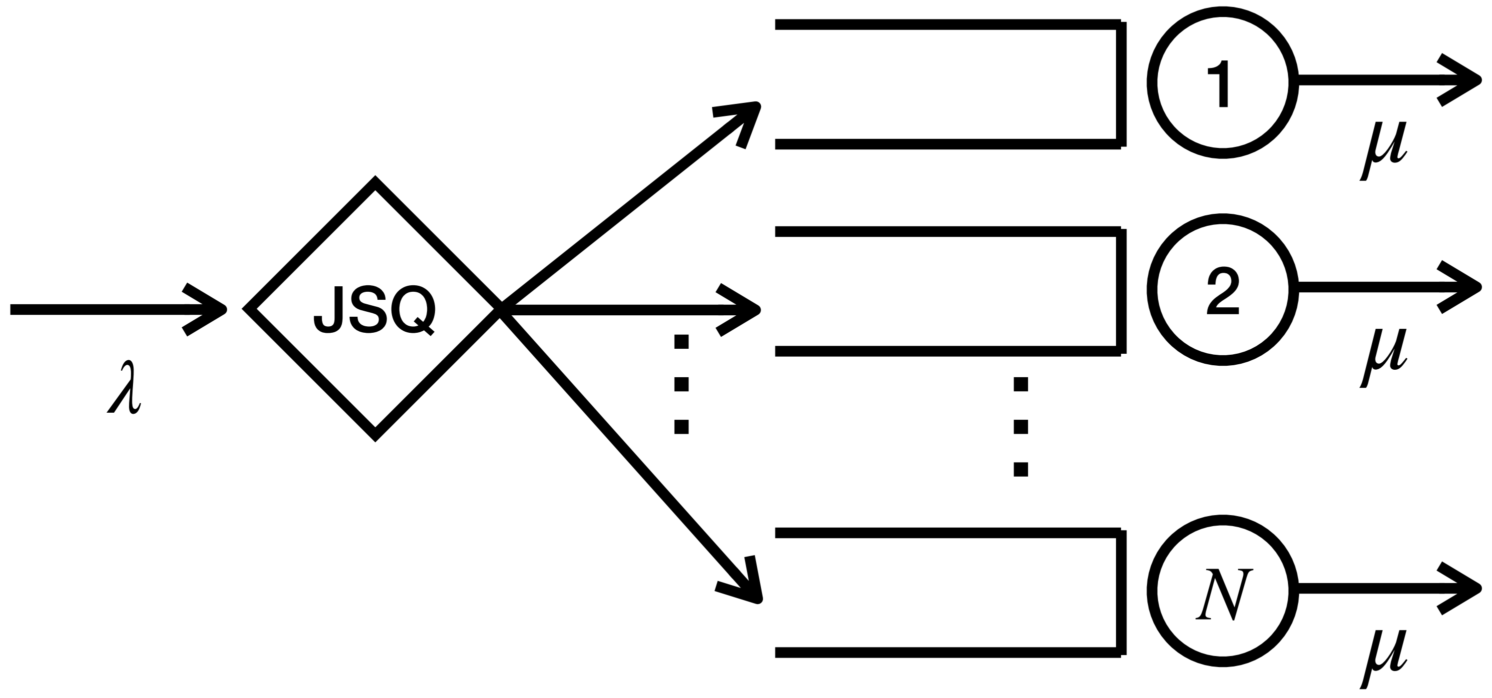
$$\approx 1_{\{\sum q_j = 0\}}$$

SSC

$$\vdots$$

$$\mathbb{E}\left[e^{\theta\epsilon \sum q_i}\right] = \frac{1}{1 - \theta N} \iff \epsilon \sum q_i \sim \text{Expo}\left(\frac{1}{N}\right)$$

Asymptotic Regimes



Large Deviations:
(a.k.a. rare events)
$\lim_{x\to\infty} P(q_i > x)$

Mean Field:
(a.k.a. many servers)
$N \to \infty$

Many-Server Heavy-Traffic:
$N\lambda = N(1 - N^{-\alpha}), \quad \mu = 1$
$\epsilon = N^{1-\alpha}, \alpha > 0$
$N \to \infty$

Heavy Traffic:
$\epsilon := N\mu - \lambda$
$\epsilon \downarrow 0$

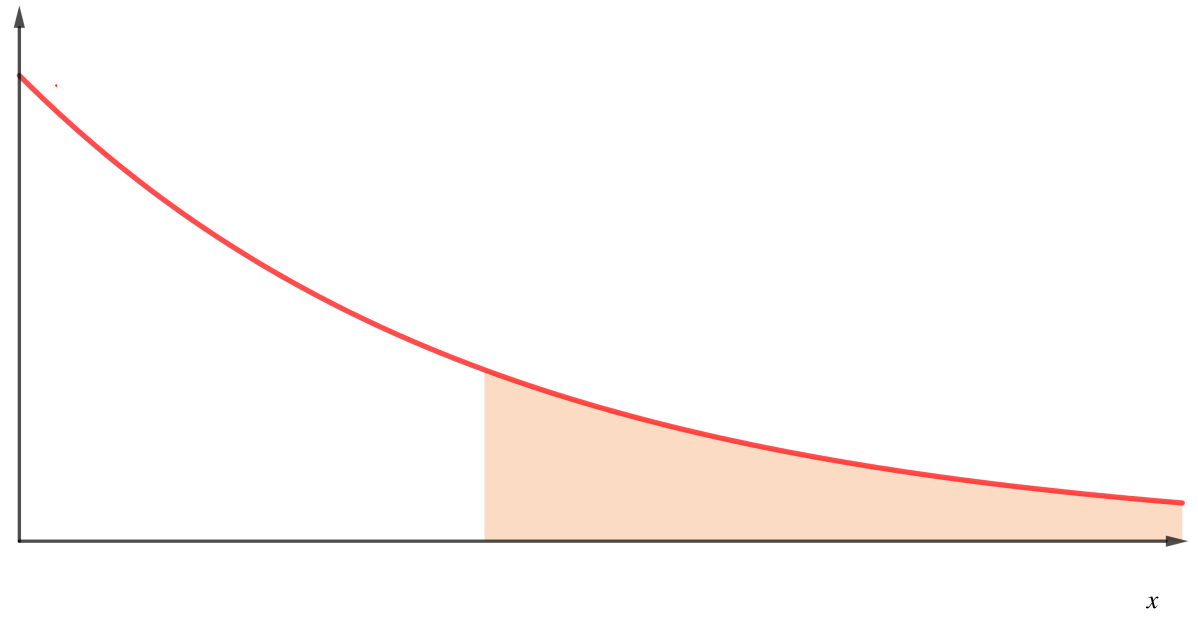
$\alpha = 0$ — $\alpha = 1$ — $\alpha > 1$

Literature:

Mukherjee et al. 2018
Mitzenmacher 2001
Liu, Ying 2020
Varma, Castro, Maguluri 2022
Eschenfeldt, Gamarnik 2018
Banerjee, Mukherjee 2019
Braverman 2020
Liu, Ying 2022
Zhao, Banerjee, Mukherjee 2021
Gupta, Walton (2019)
…

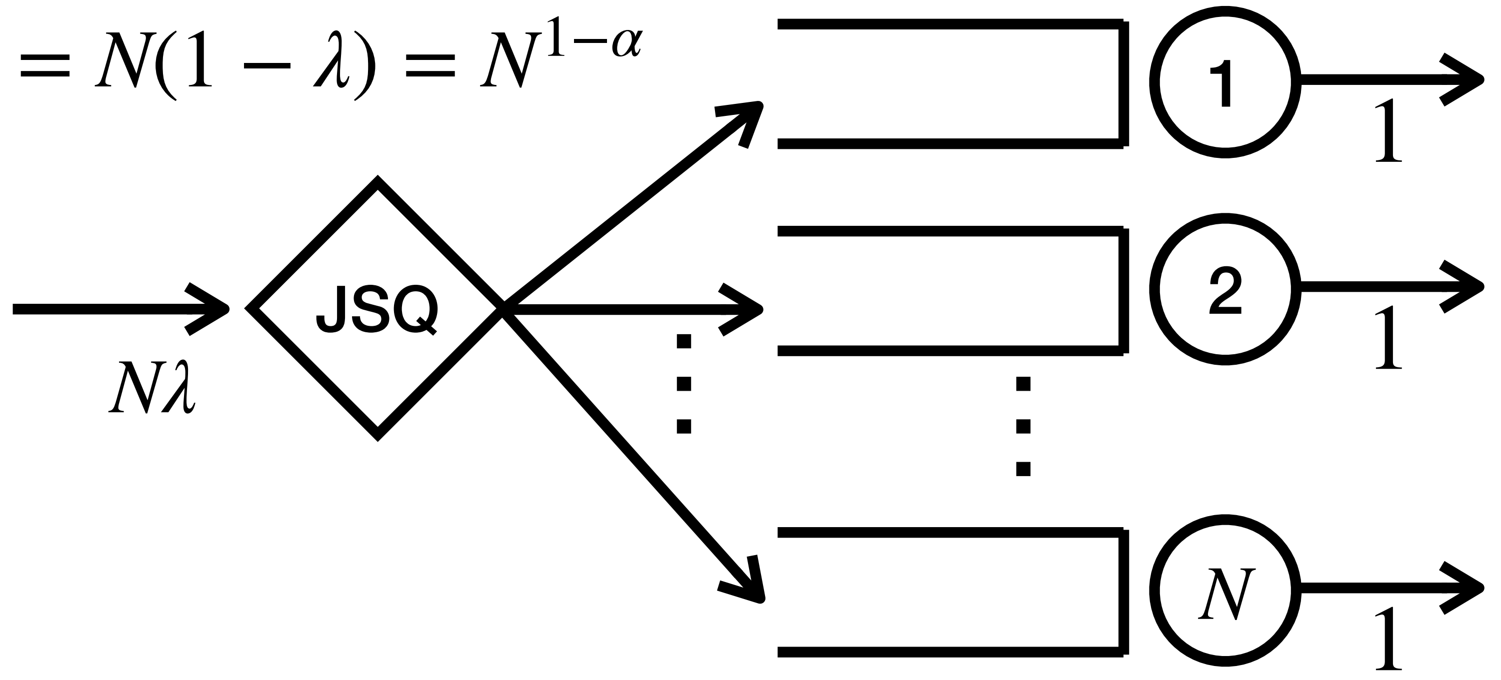
[Jhujhunwala, HL, Maguluri, '23]

Characterize $\mathbb{P}(q_i > x)$



Non-Asymptotic JSQ

$\epsilon_N = N(1-\lambda) = N^{1-\alpha}$



Theorem: [Jhunjhunwala, HL, Maguluri, '23]

For N sufficiently large, and $\theta_N = N^\alpha \log\left(\frac{1}{1-N^{1-\alpha}}\right)$

$$e^{-\theta_N x} \leq \mathbb{P}\left[\epsilon_N \sum_{i=1}^{N} \frac{q_i}{n} > x\right] \leq e^{-\theta_N x}\left(1 - \kappa N \epsilon_N \log \epsilon_N\right) 2ex$$

Large deviations regime:

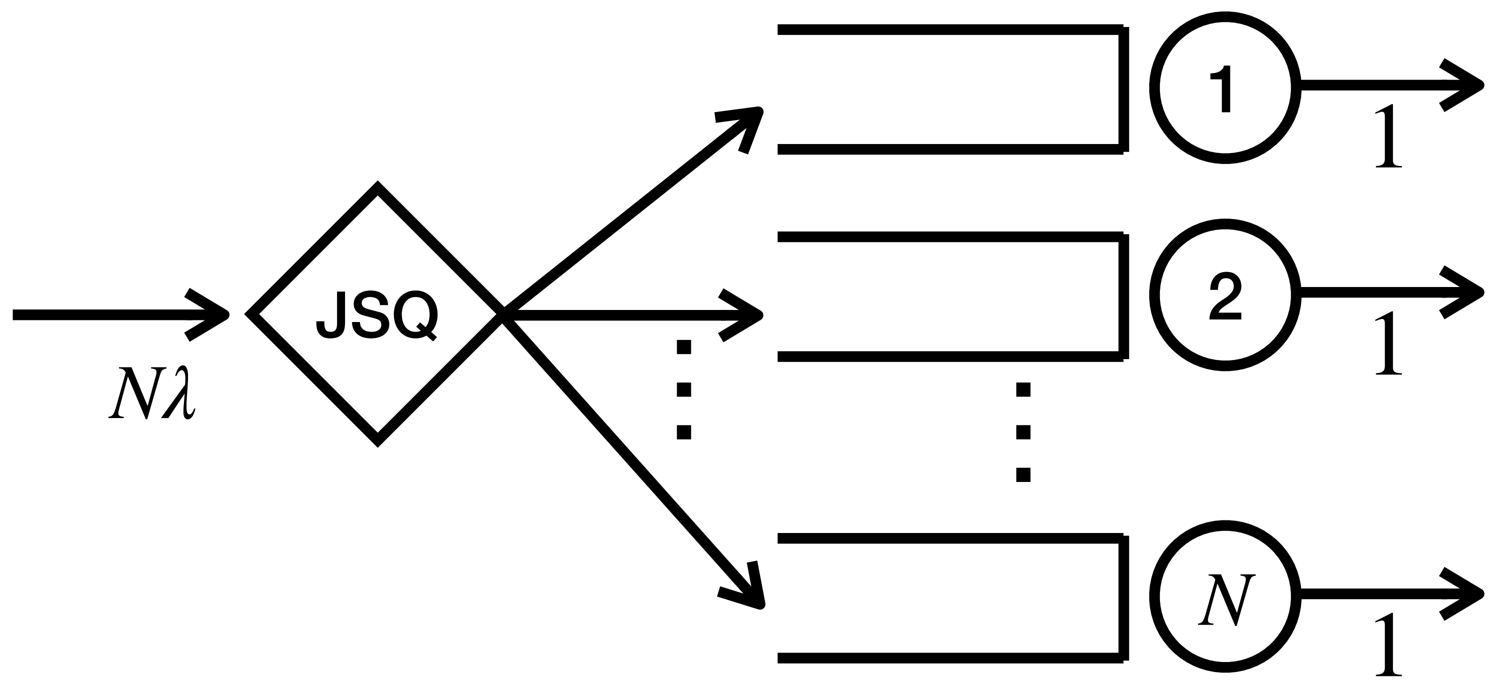
$$\lim_{x\to\infty} -\frac{1}{x}\mathbb{P}(\cdot) = \theta_N$$

State Space Collapse Violation

Pre-limit error

Non-Asymptotic Analysis

Refined SSC



Proposition: [Jhunjhunwala, HL, Maguluri, '23]

For large N, we have

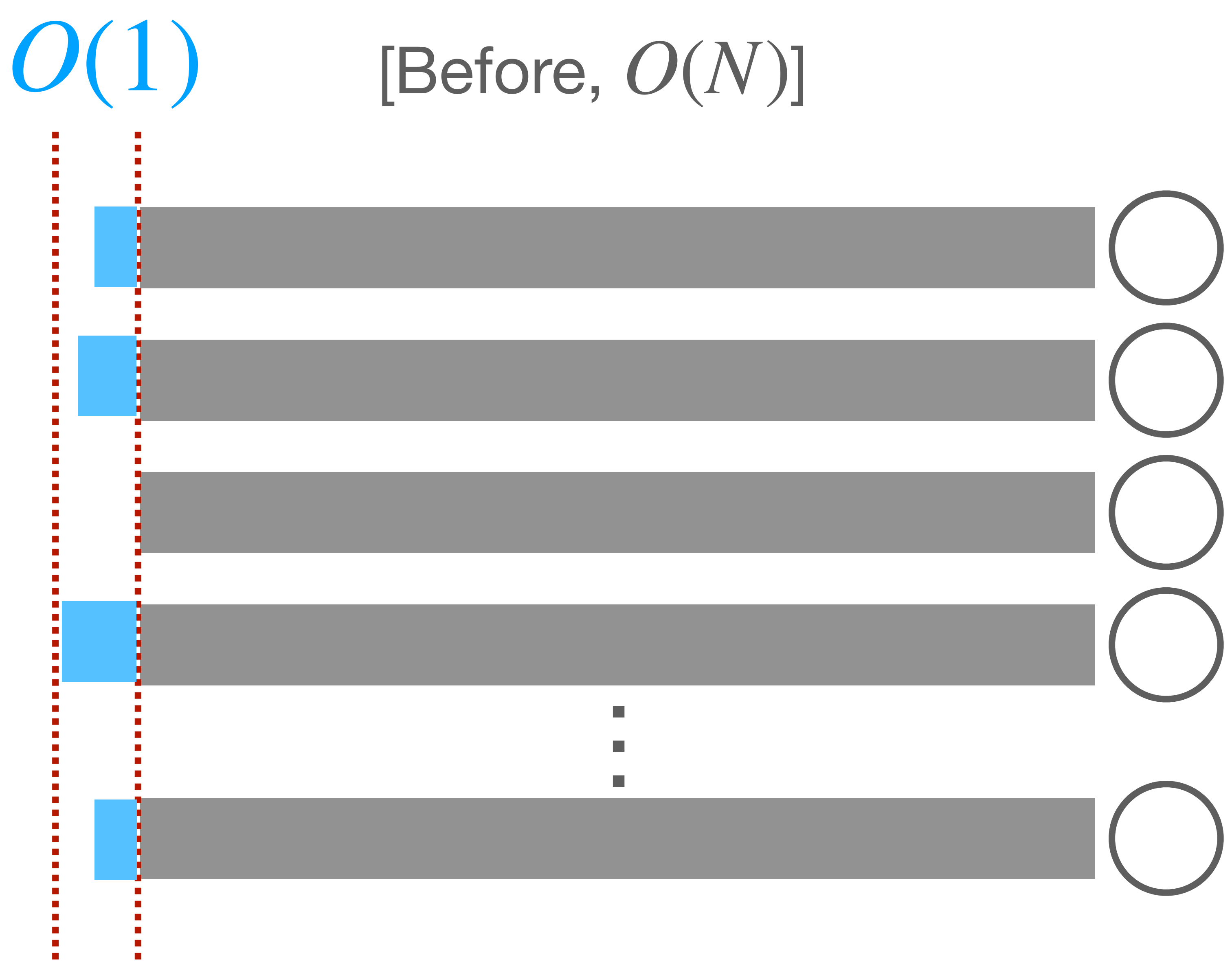
$$MGF(q_{SSQ}) \leq MGF\left(\frac{1}{N}\sum q_i\right) \leq \left(1 + \kappa N^{1-\alpha} \log N\right) MGF(q_{SSQ})$$

State Space Collapse Violation

N large

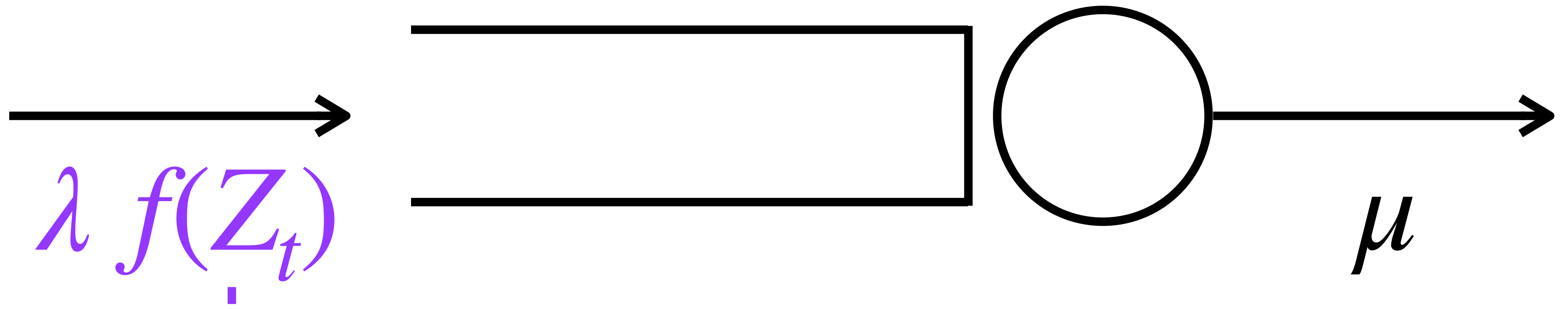
$\iff$ $\epsilon = N^{1-\alpha}$ small (recall $\alpha > 1$)

$\implies$ Heavy traffic behavior

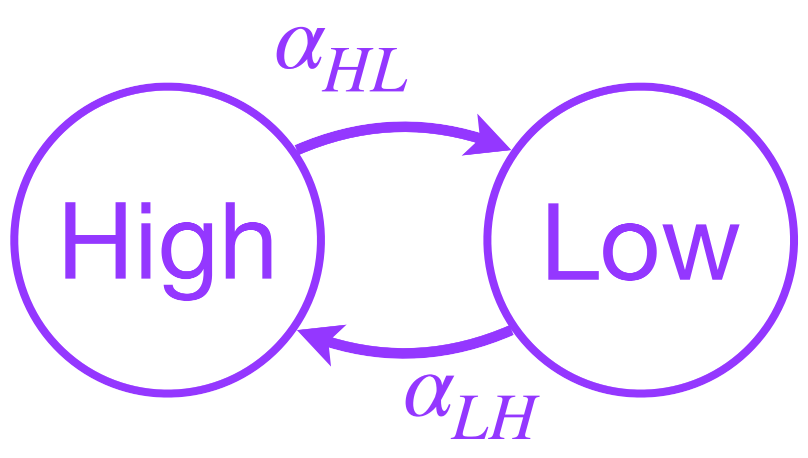


Markov-Modulated Arrivals

Ongoing Work



$\{Z_t : \ t \geq 0\}$ is a CTMC



- Arrival rate depends on Z_t
- $\{q_t : \ t \geq 0\}$ is **<u>not</u>** a CTMC
- $\{(q_t, Z_t) : \ t \geq 0\}$ is a CTMC

Falin, Falin (1999):

- Workload distribution of Markov$/G/1$ queue
- Methodology:
  Forward eqns + Laplace transform
  + Properties of Q matrix
- Hard to generalize to multiple servers/queues

- **Transform method:**

Step 1: Drift of exponential test function

$V_\theta(q) \cancel{=} e^{\theta q}$

$\{q_t\}$ is not a CTMC

$V_\theta(q, z) \cancel{=} e^{\theta q} \Big| Z = z$

$\Delta V_\theta(q, i)$ doesn't have simple terms

$V_\theta(q, z) = e^{\theta q}\, g_\theta(z) \Big| Z = z$

where $g_\theta(z)$ is such that the drift is "easy"

Transform Method for Markov$/M/1$

Step 1: Drift of exponential test function

$$V_\theta(q, H) = e^{\theta q}\left(\lambda f(L)(e^{\theta} - 1) + \mu(e^{-\theta} - 1) - \alpha_{LH} - \alpha_{HL}\right) \Big| Z = H$$

$$V_\theta(q, L) = e^{\theta q}\left(\lambda f(H)(e^{\theta} - 1) + \mu(e^{-\theta} - 1) - \alpha_{LH} - \alpha_{HL}\right) \Big| Z = L$$

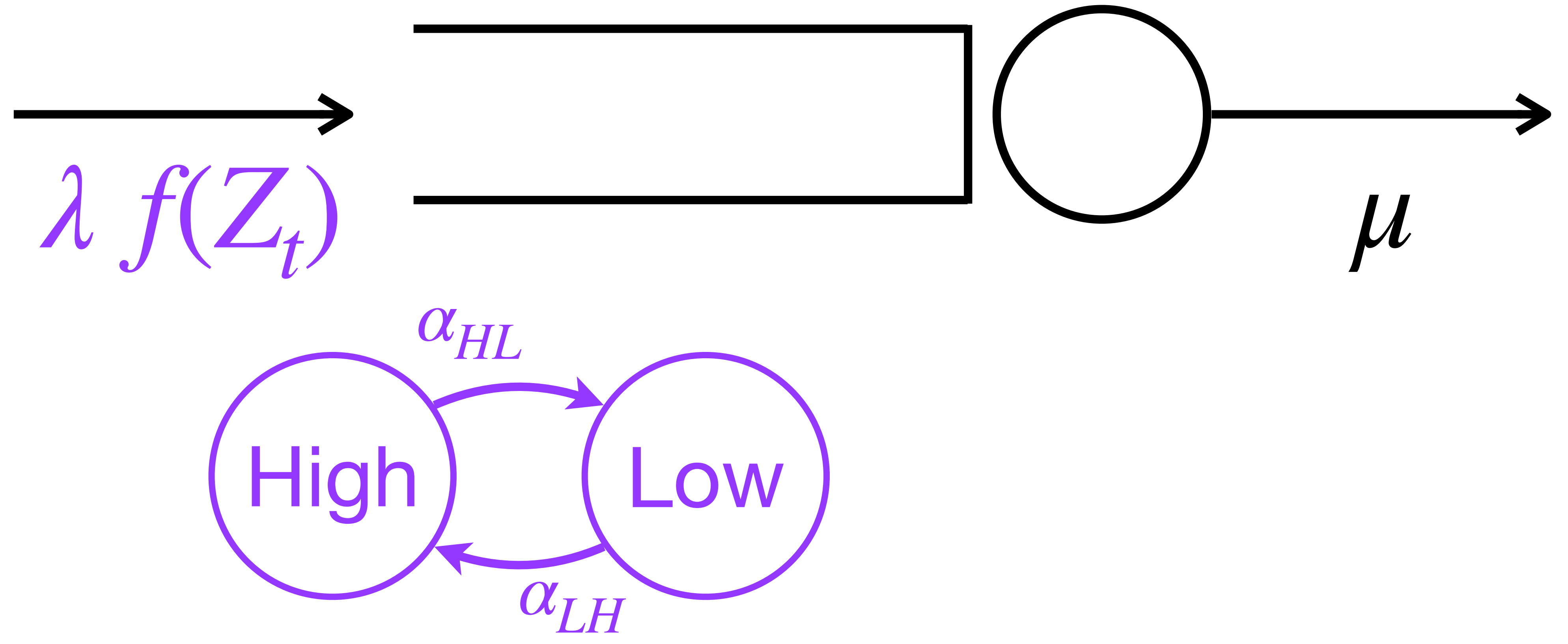
Step 2: Set drift to zero

$\vdots$

Step 3: Set $\theta = 0$

$\vdots$

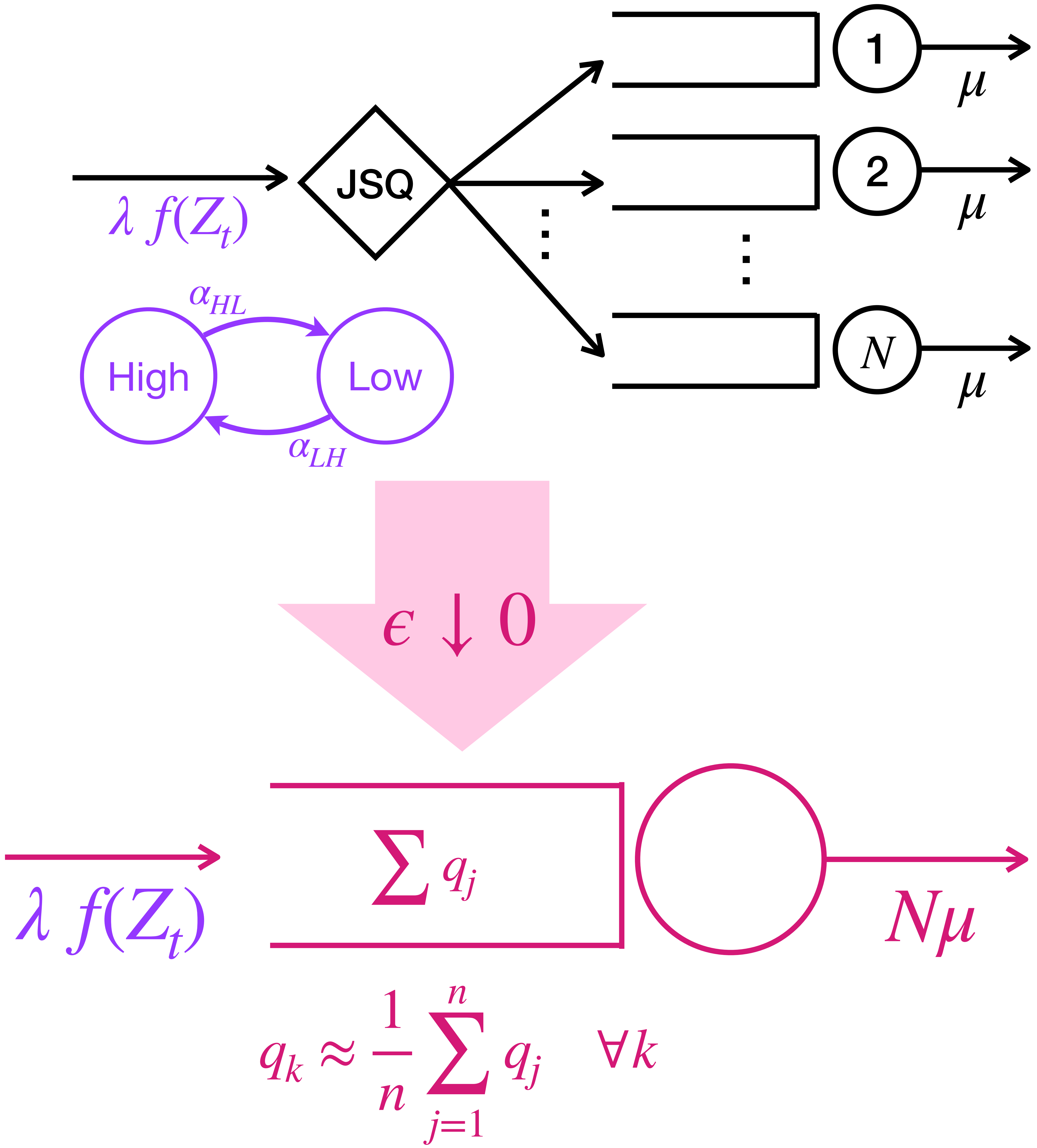
$\mathbb{E}\left[e^{\theta q}\right]$ = Exact expression in terms of $\lambda, f(H), f(L), \alpha_{HL}, \alpha_{LH}, \mu$



JSQ with Markov-Modulated Arrivals

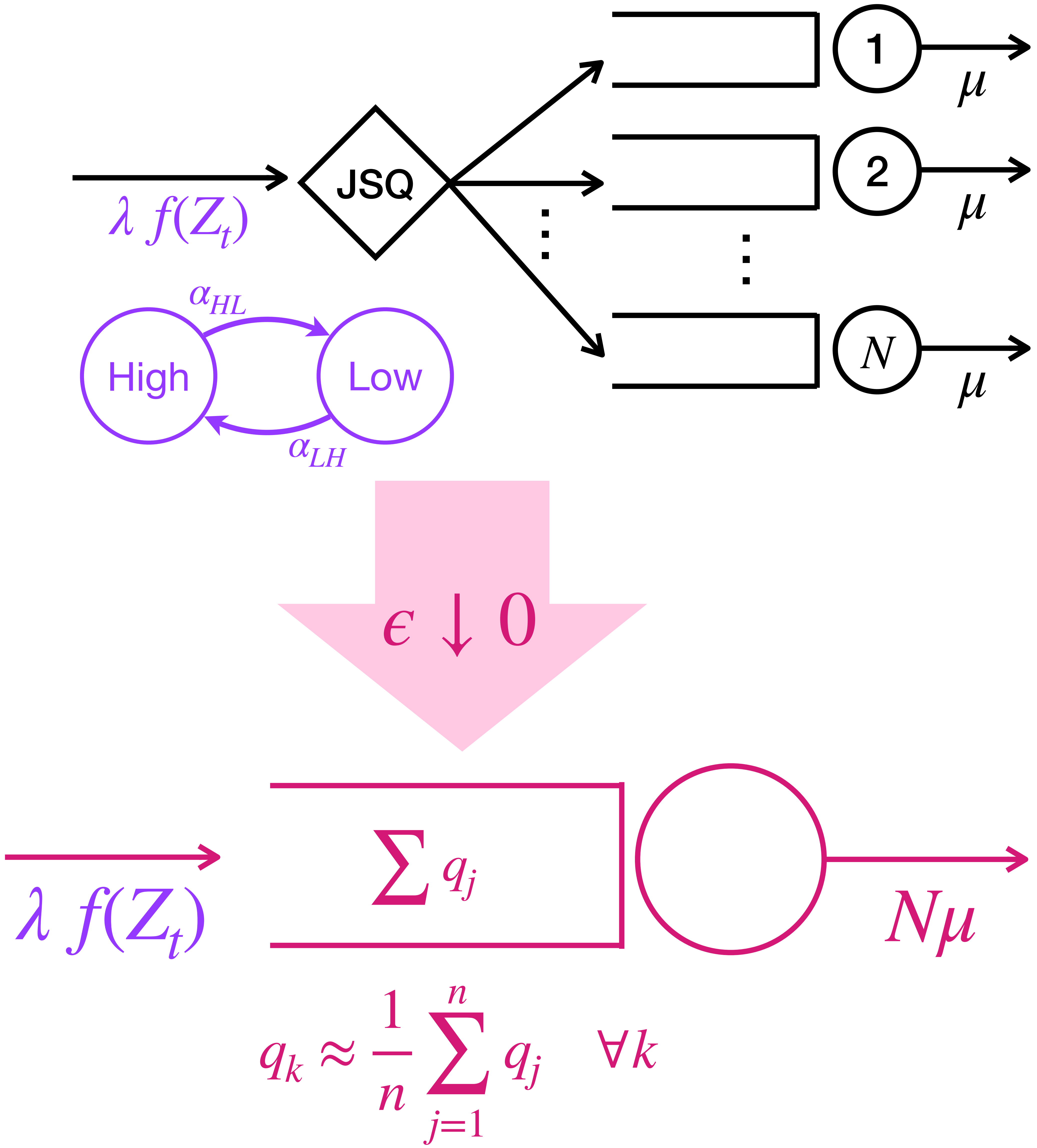
Heavy-Traffic Analysis:

- Load: $\epsilon = n\mu - \lambda_{avg}$
  - $\lambda_{avg} = \lambda(\pi_L f(L) + \pi_H f(H))$
  - $\boldsymbol{\pi}$ is stationary distribution
- In heavy traffic, $\epsilon \downarrow 0$ and system experiences Complete Resource Pooling (CRP)



Key Takeaways

- **Transform Method:**
  - Tractable expressions to compute distributions
  - Easily incorporate CRP condition in heavy traffic
  - Test function is key
  - Generalize to any CRP system
- **What is a good test function?**
  - Capture dynamics of queueing system
  - Represent state-space collapse
  - Yields tractable drift
- **Future work:**
  - General Markov-modulation process
  - Markov-modulated service rate



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Expected Delay	➡ Systems with multiple bottlenecks ➡ Computing the mean
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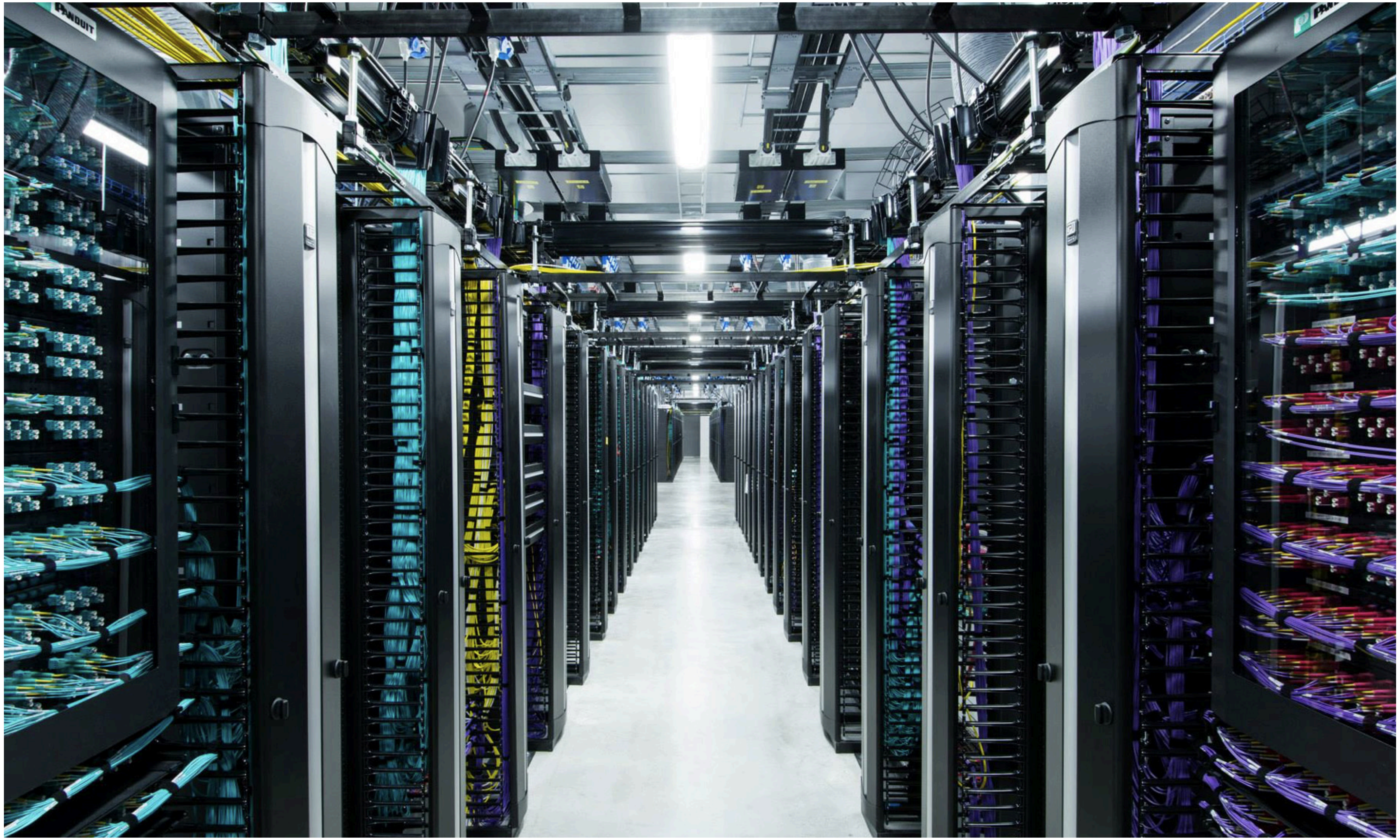
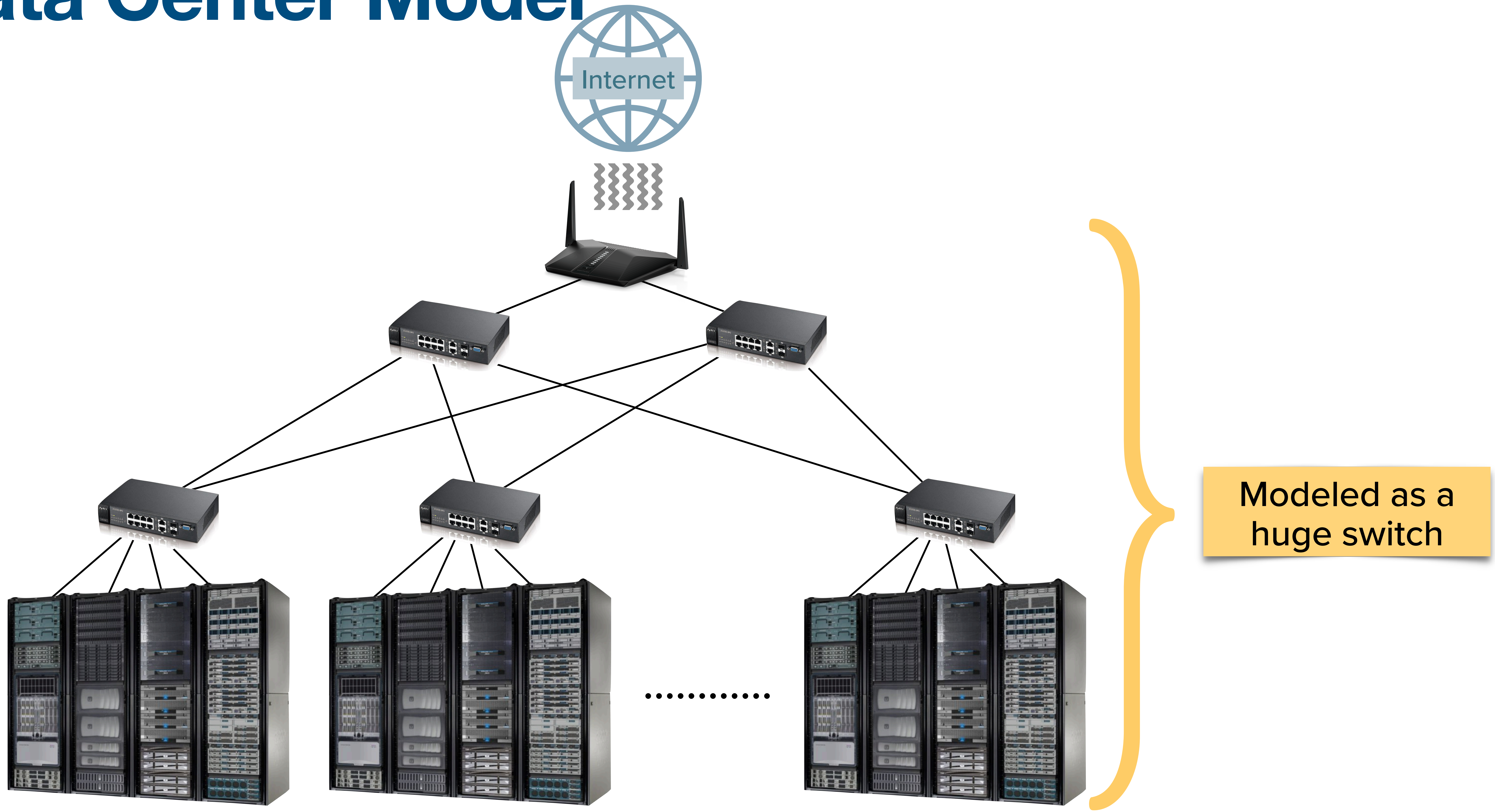
Data Centers



Data Center Model



Input Queued Switch

- Discrete time model

- n input ports and n output ports

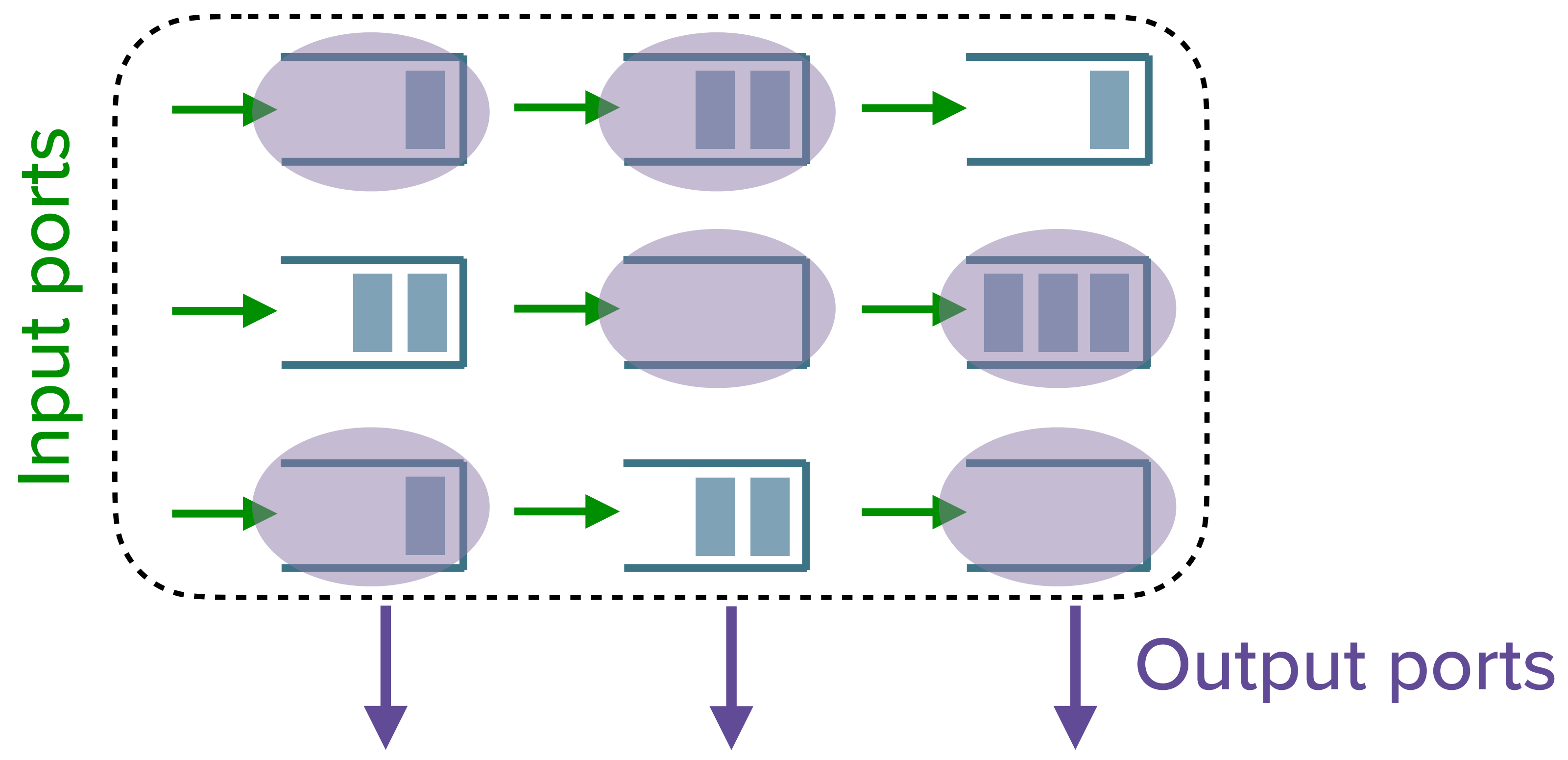
  Jobs arrive at input ports, and go to the desired output port

  Jobs are processed at output ports and take exactly one time slot

- **Constraint:** At most **one** job can be processed from each input port and at each output port

- **Scheduling:**

  Which queues to serve?
  MaxWeight: Largest total queue length



Heavy-traffic analysis:

- Load system close to its maximum capacity
- Define
  $\epsilon := 1 -$ Arrival rate to each row/column
- Take $\epsilon \downarrow 0$

State Space Collapse (SSC):

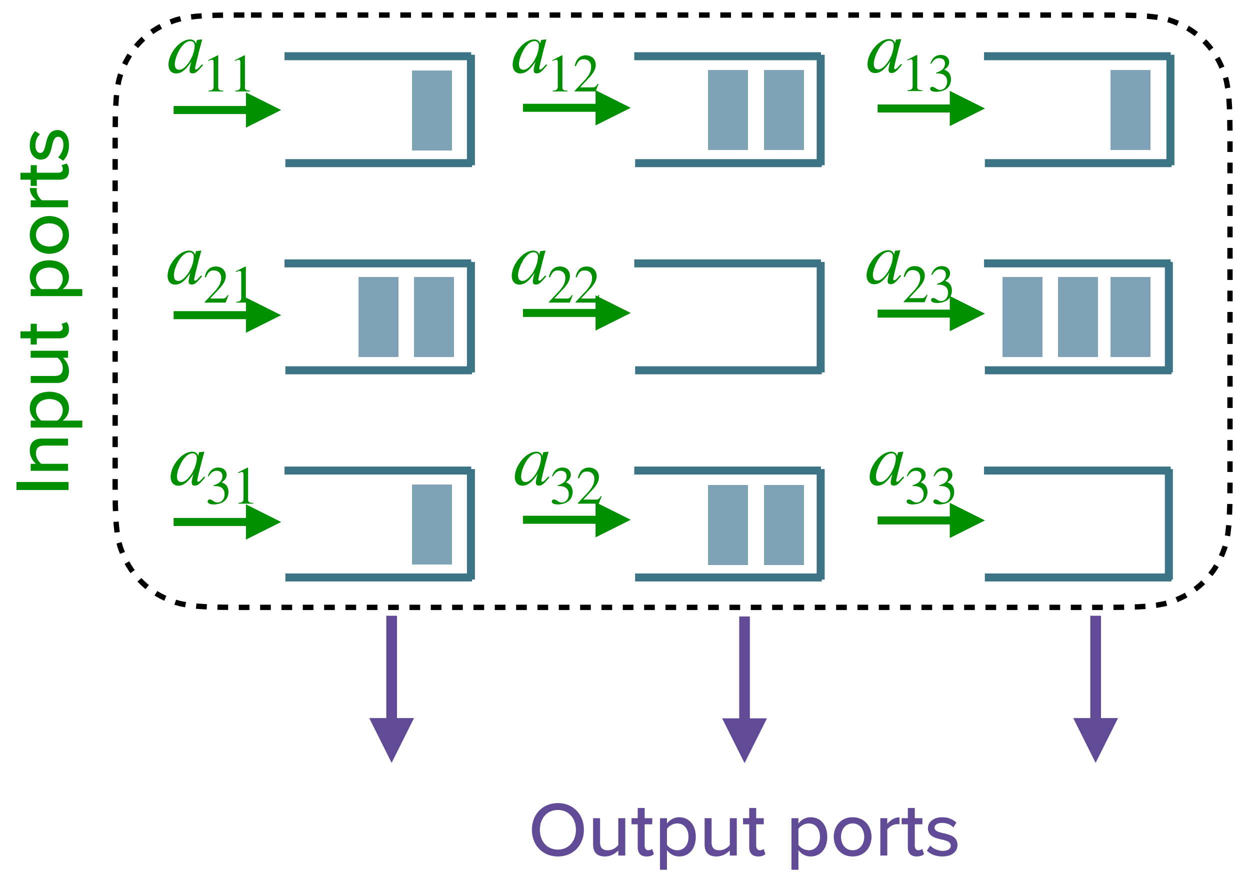
- All schedules have the same # jobs
- Describes a subspace of dimension $2n - 1$

Performance Analysis of the Switch

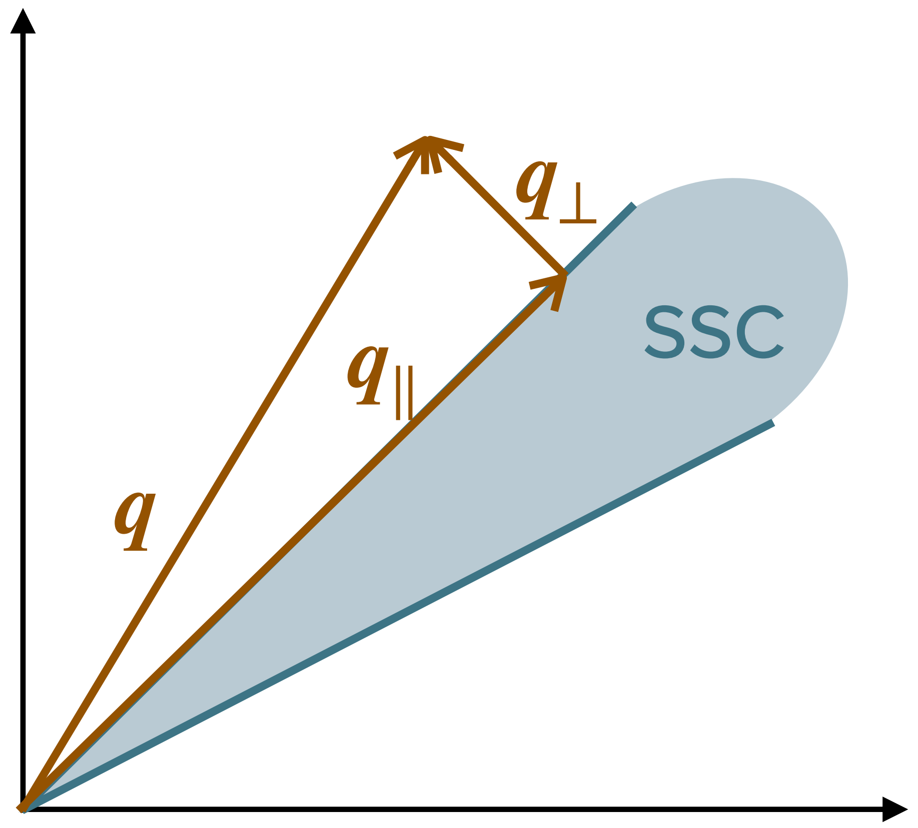
Theorem: [HL, Maguluri, '21]

In an $n \times n$ input-queued switch,

$$\lim_{\epsilon \downarrow 0} \epsilon \mathbb{E}\left[\sum_{i,j} q_{i,j}\right] = \frac{n}{2} \sum_{ij} \left[\underbrace{\underbrace{\sum_{i'} \frac{Cov[a_{ij}, a_{i'j}]}{n}}_{\text{Row average}} + \underbrace{\sum_{j'} \frac{Cov[a_{ij}, a_{ij'}]}{n}}_{\text{Column average}} - \underbrace{\sum_{i'j'} \frac{Cov[a_{ij}, a_{i'j'}]}{n^2}}_{\text{Total average}}}_{\text{Subspace describing SSC}} \right]$$



Multi-Dimensional SSC



✓ We characterize the mean queue length
How? Study $\|\boldsymbol{q}_{\parallel}\|^2$

✓ Result is valid for:

- ✓ General customer and server types
- ✓ Any dependence constraints among servers
- ✓ These constraints can change over time

✓ Higher moments cannot be computed with current tools

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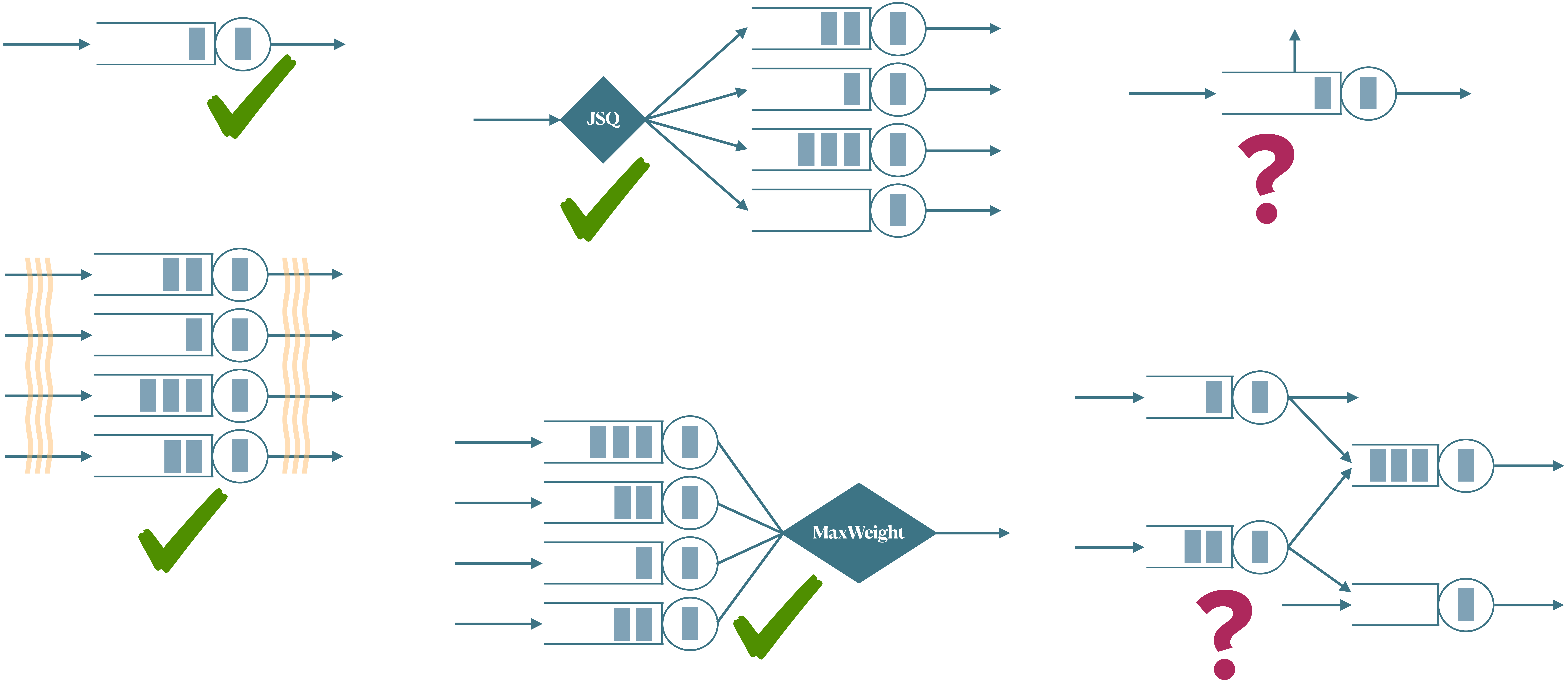
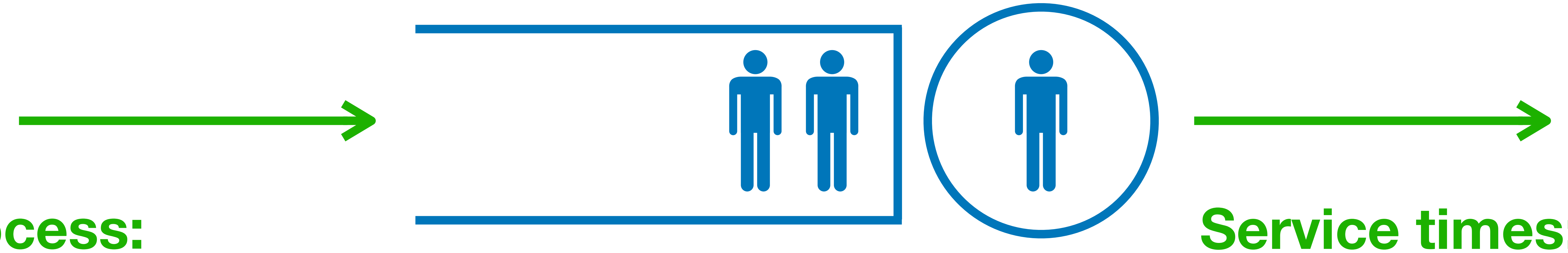
My Research So Far



Transient Analysis

Ongoing Work



Arrival process:

- Poisson
- Renewal
- Markov-modulated
- Cyclic

Asymptotic regimes:

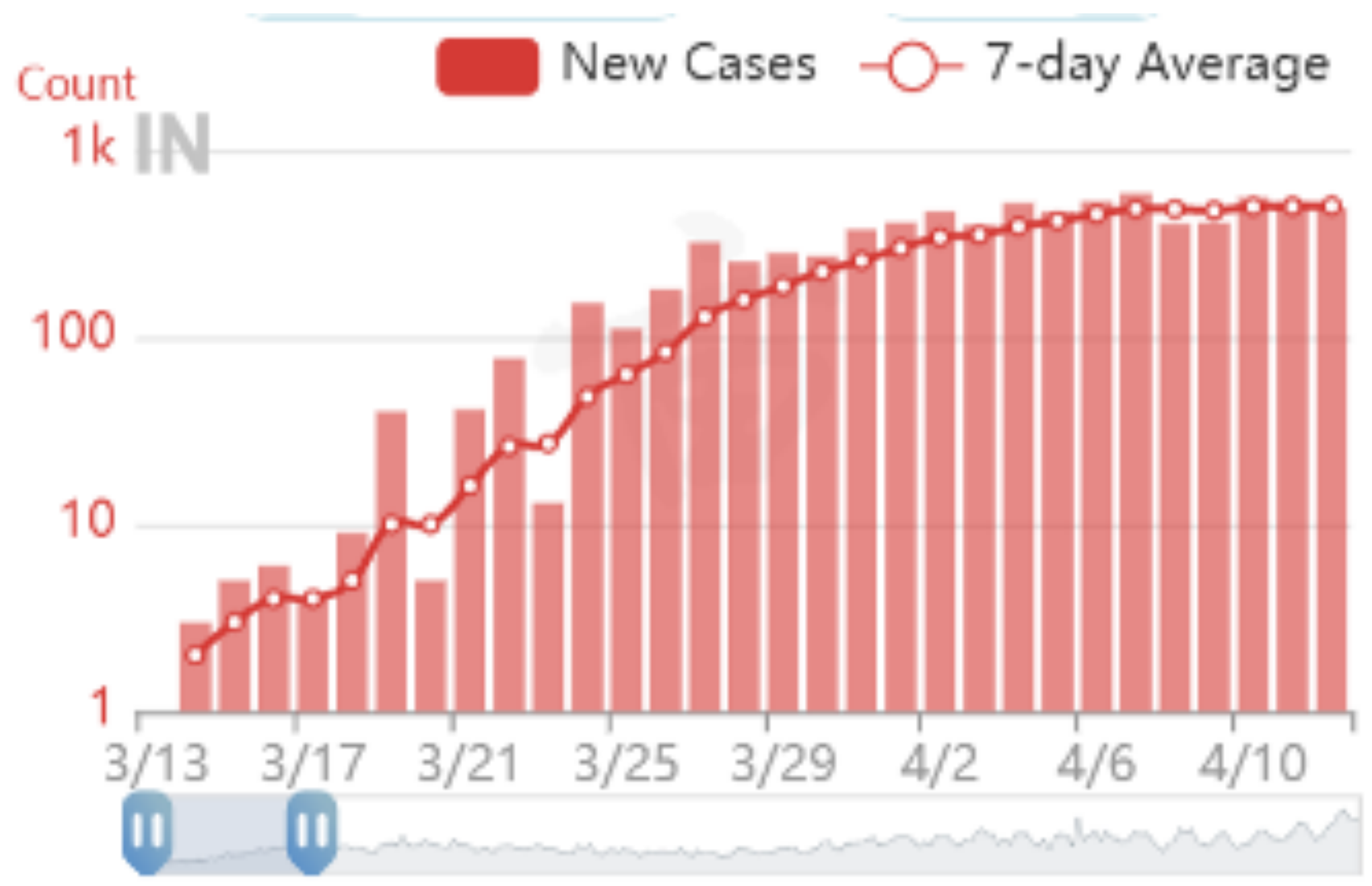
- Heavy traffic
- Mean field
- Many-server heavy-traffic
- Steady state

Service times:

- Exponential
- General distribution
- …

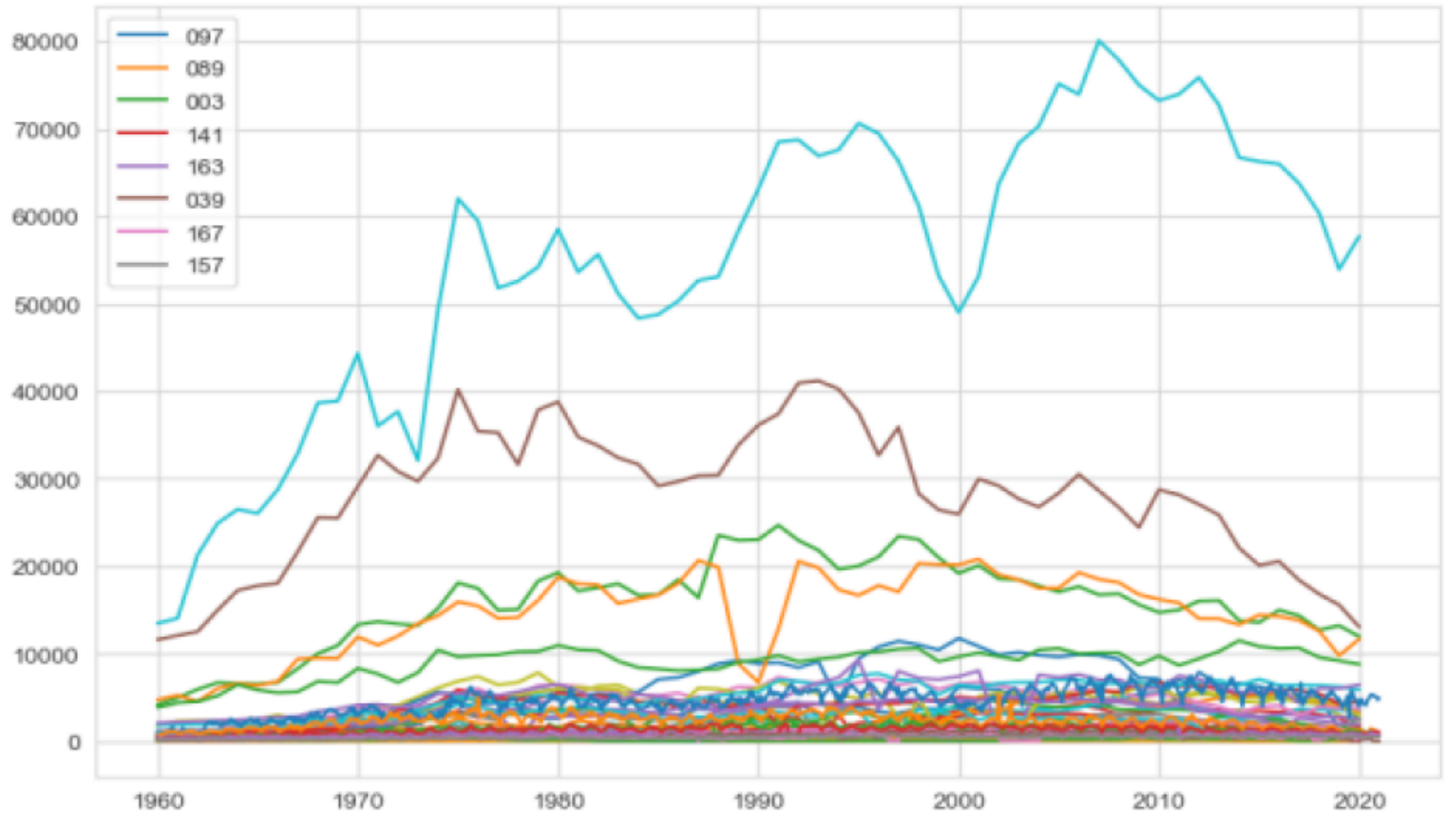
Arrival Process in Real-Life Systems

Covid-19 cases in Indiana (2020)



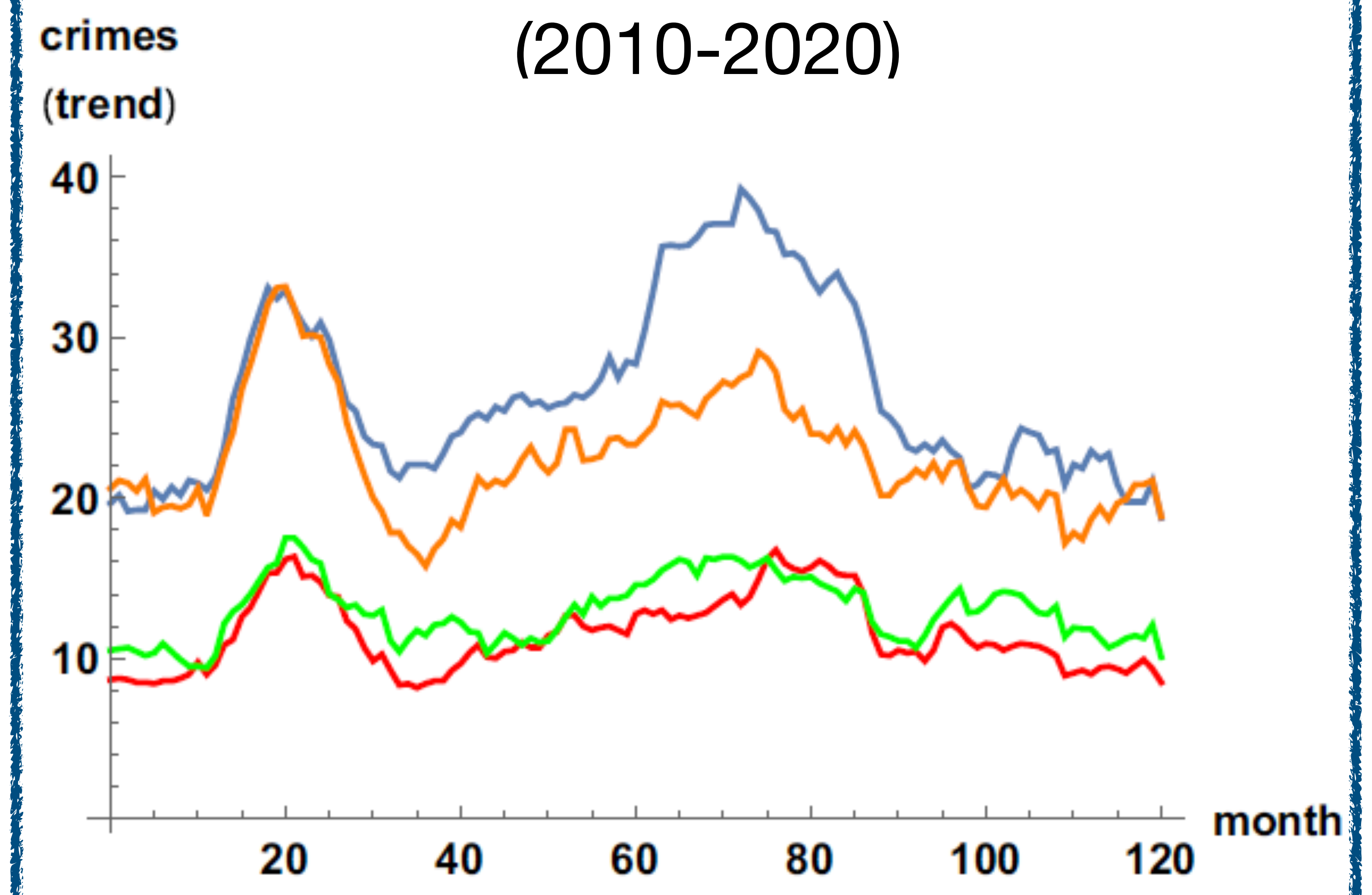
Source: https://coronavirus.1point3acres.com/

Yearly crime incidents in Indiana



Source: Indiana police department

Criminal incidents in Valencia (2010-2020)



Source: Calatayud et.al. (2023)

Arrival rate modeled as Cox process

Key challenges:

- Nonstationary
- Noise around "main trend"

How to compute the transient distribution (pmf) of number-in-system?

Cox process

Definition

- Doubly stochastic Poisson process

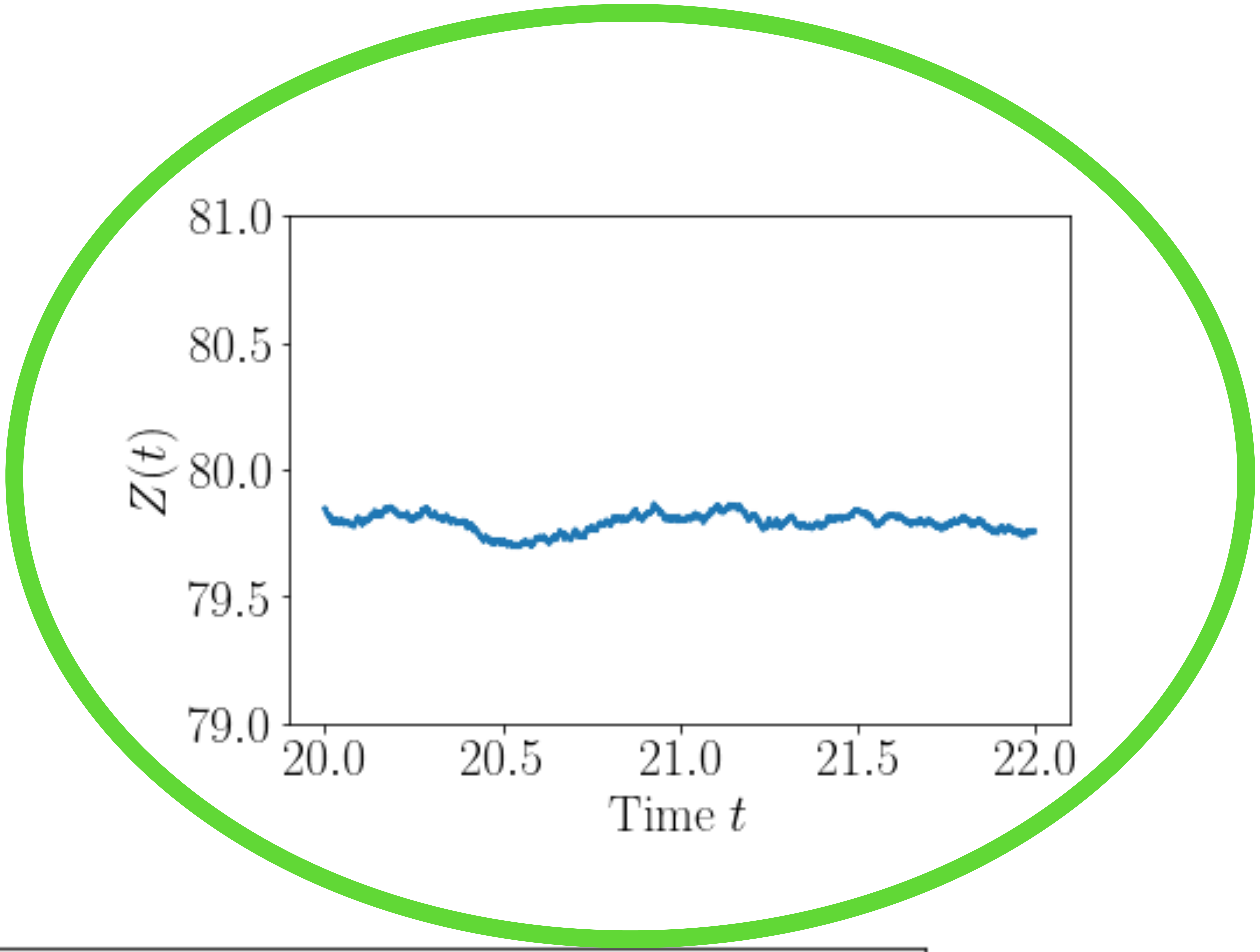
- Arrival rate process $\{Z_t : t \geq 0\}$:

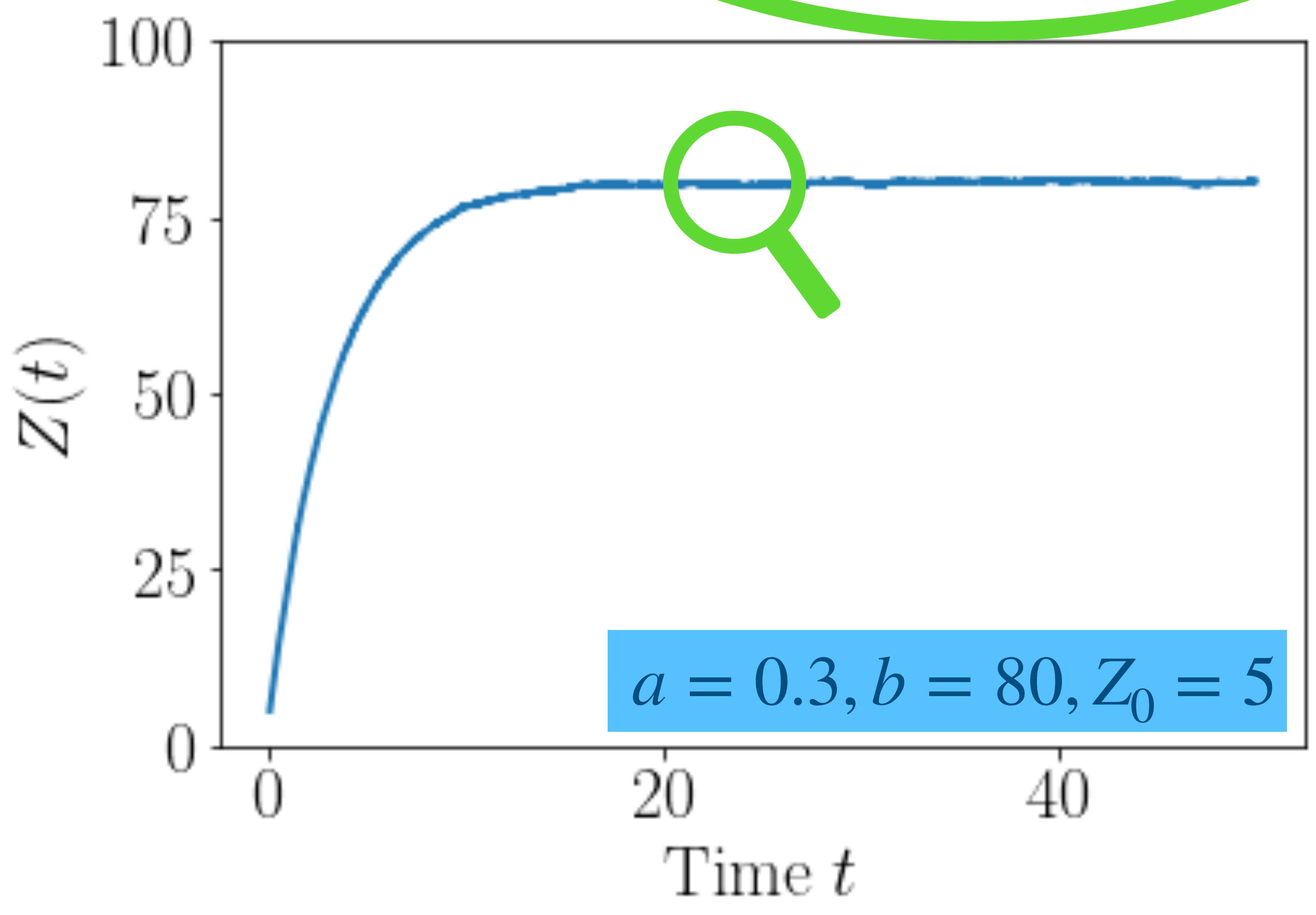
$$dZ_t = a(b - Z_t)\,dt + \sqrt{Z_t}\,dW_t$$

Convergence rate to steady state

Mean of stationary process

Wiener process (noise)





Efficient Algorithm

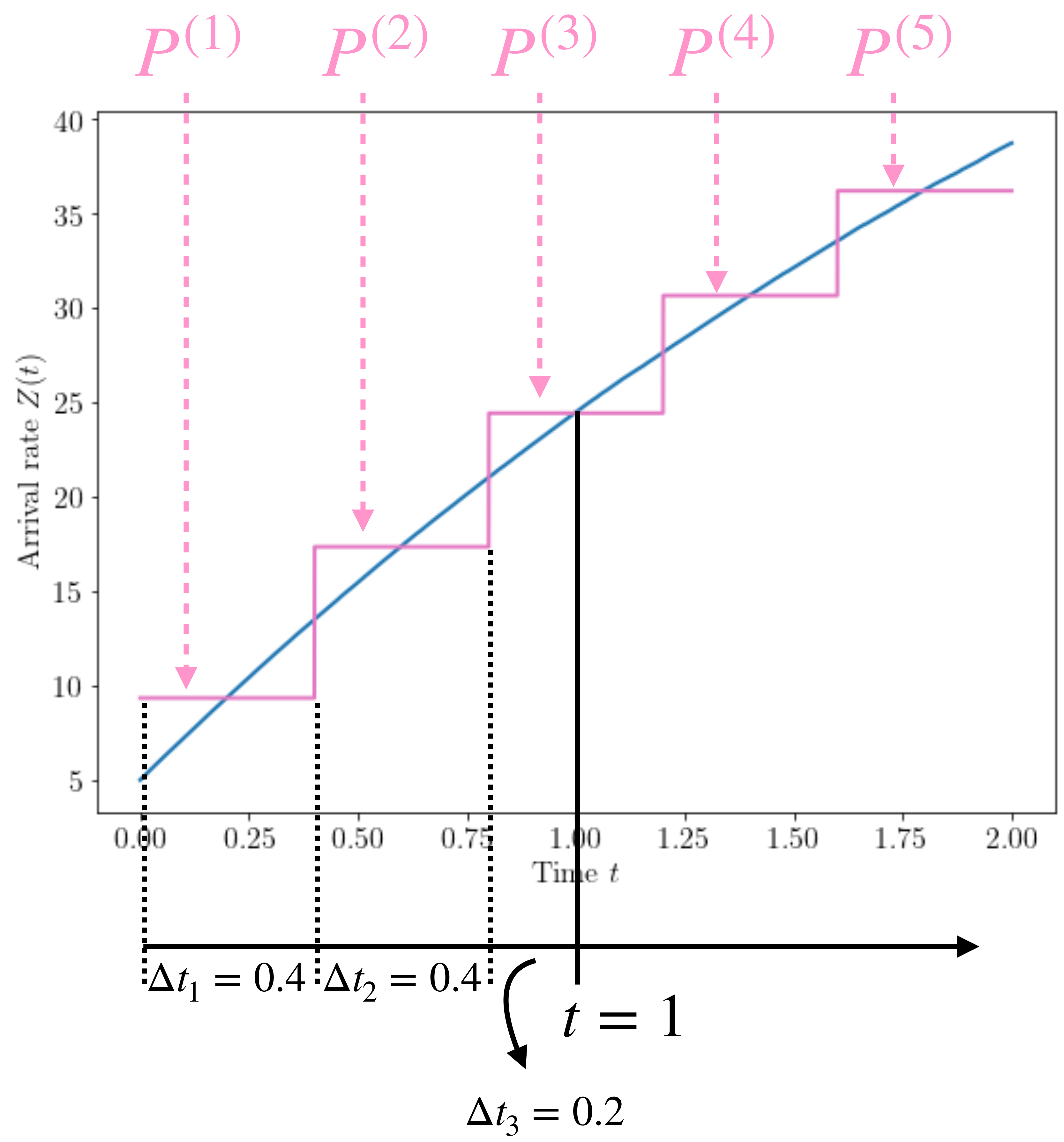
Step 1: Step function to approximate Z_t

1. Divide horizon T in intervals
2. Compute average rate in each interval

Step 2: Uniformization in each step

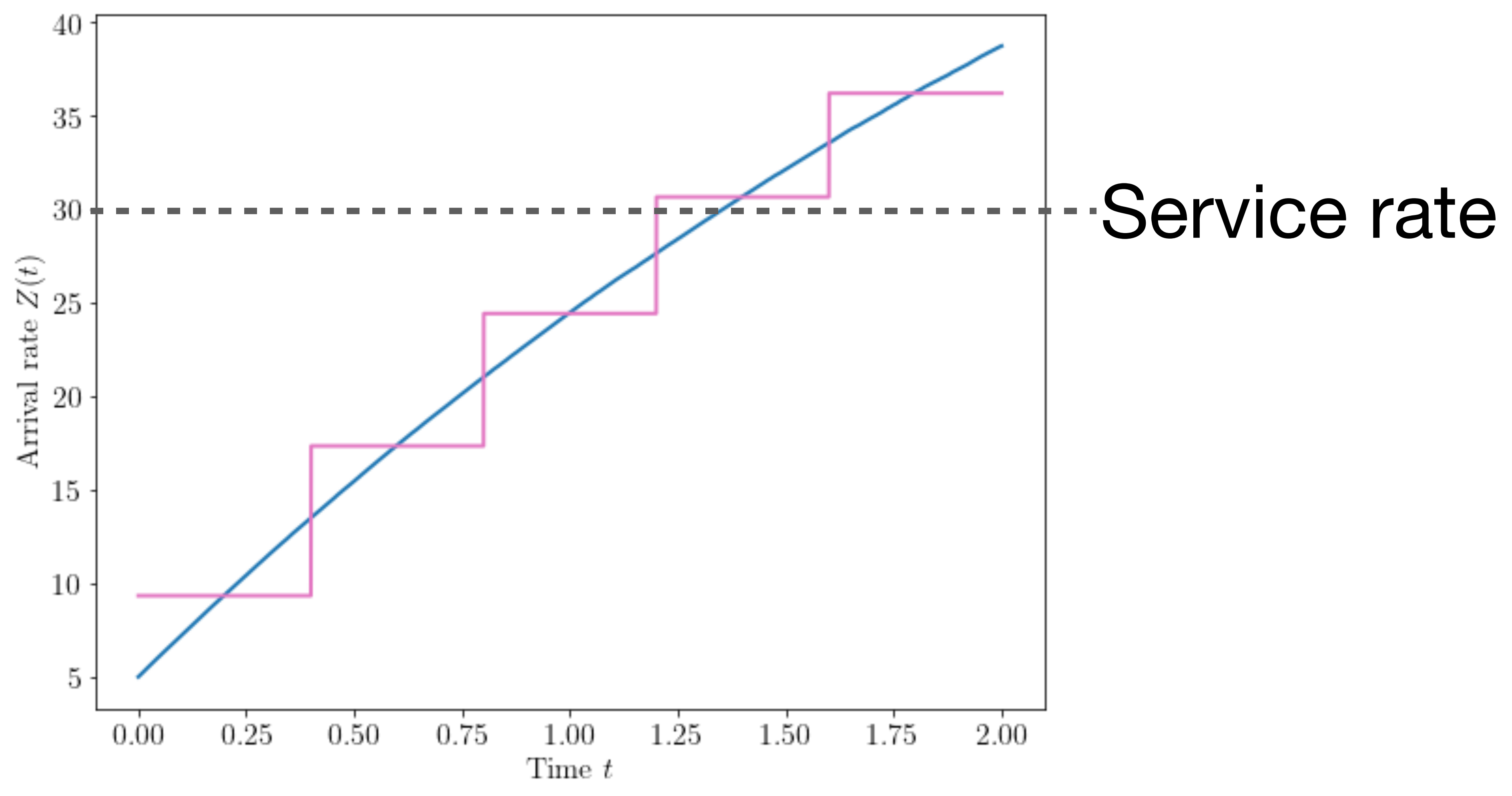
1. Compute the transition matrix in each "step"
2. Uniformization in each "step":

$$\underbrace{\boldsymbol{p}(t)}_{\substack{\text{pmf:}\\ p_j(t) = P[X(t)=j] \\ X(t) = \#\text{ in system}}} = \left(\prod_{k=1}^{\overbrace{\kappa(t)}^{\#\text{ steps in }(0,t)}} \sum_{a=0}^{\infty} \underbrace{P(a \text{ events in } \overbrace{\Delta t_k}^{\text{Time in step } k})}_{\text{Poisson pmf}} \left[P^{(k)}\right]^a \right)' \boldsymbol{p}(0)$$

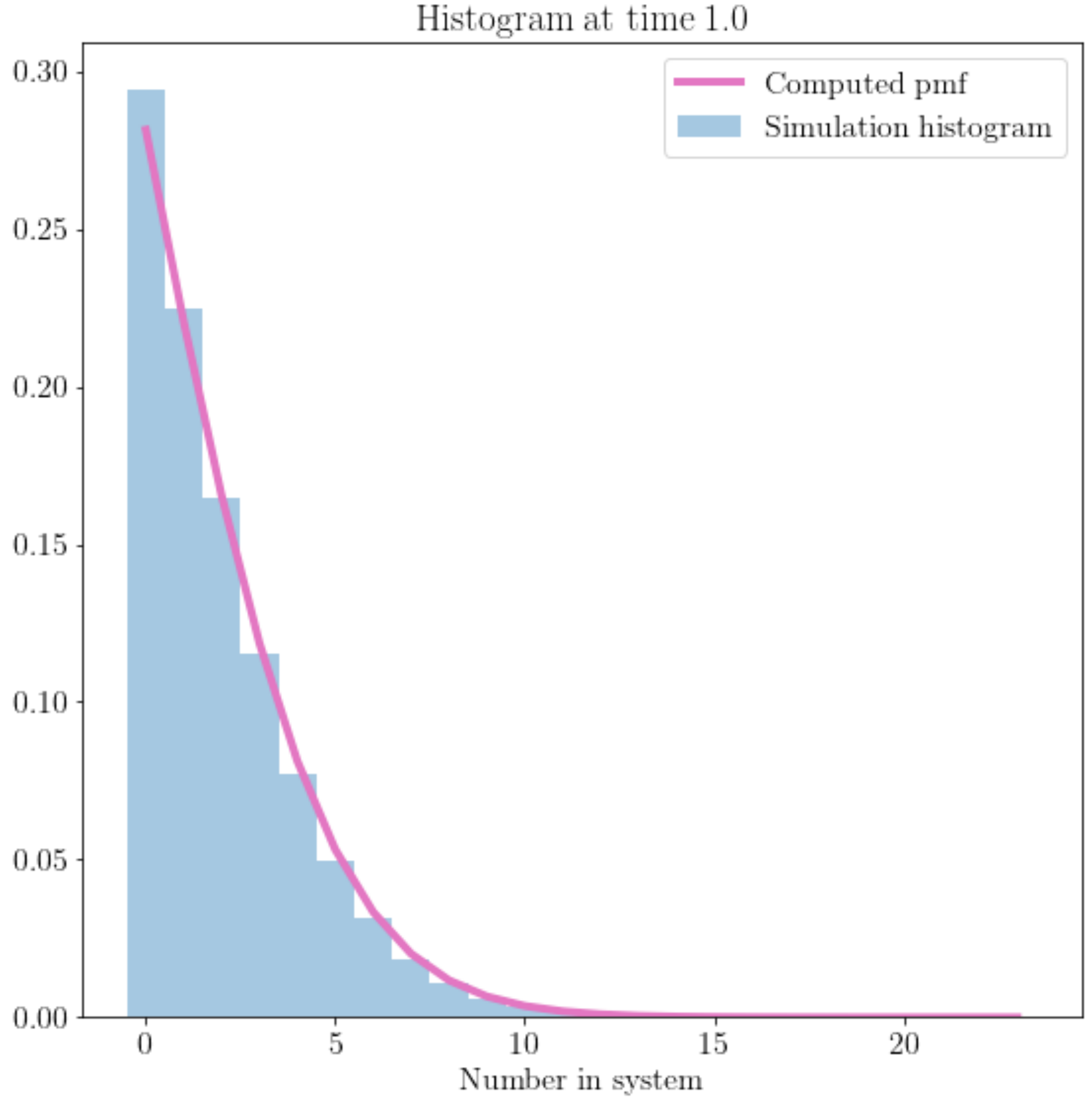


$\kappa(1) = 3$

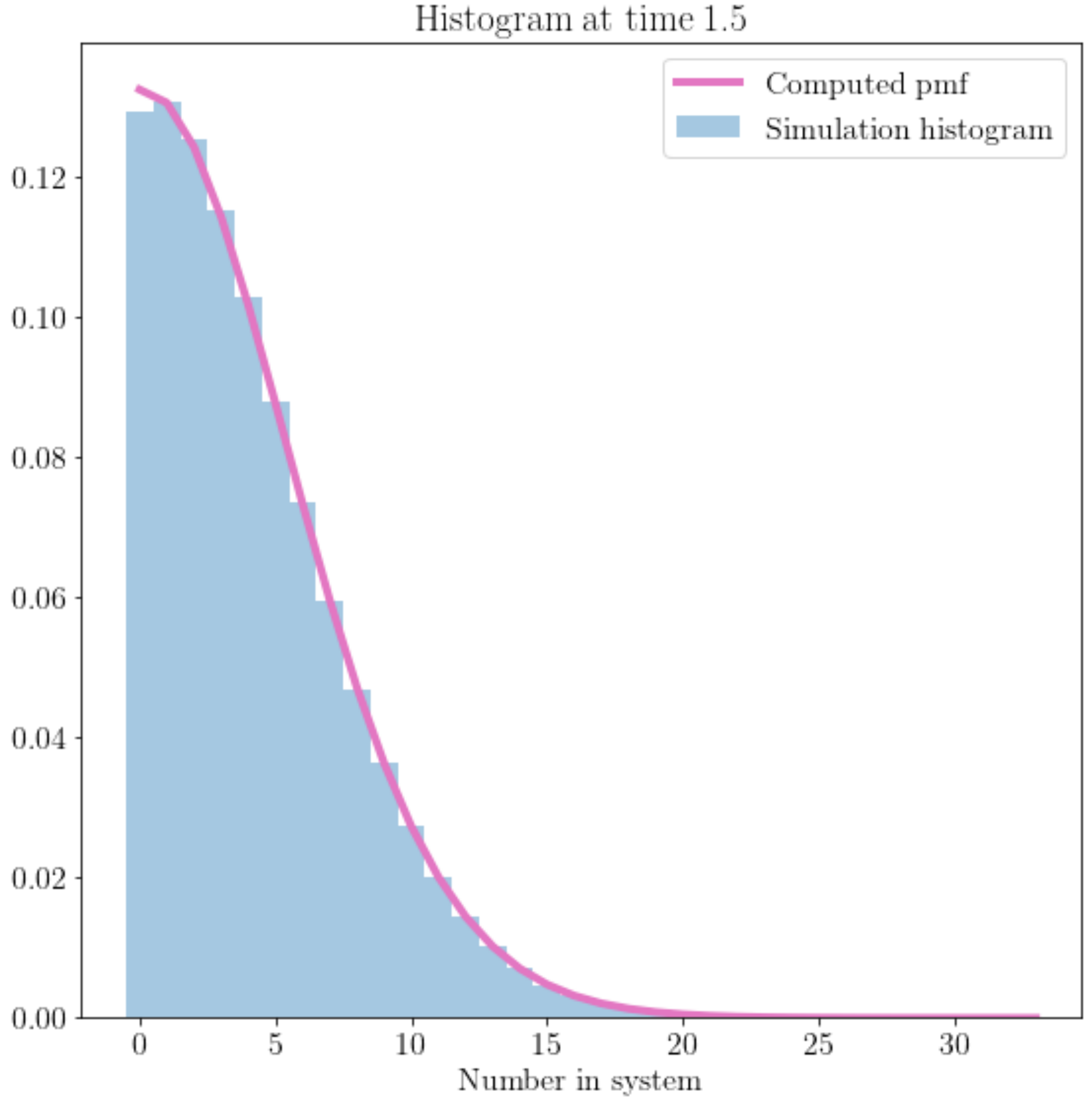
Results $Cox/M/1$



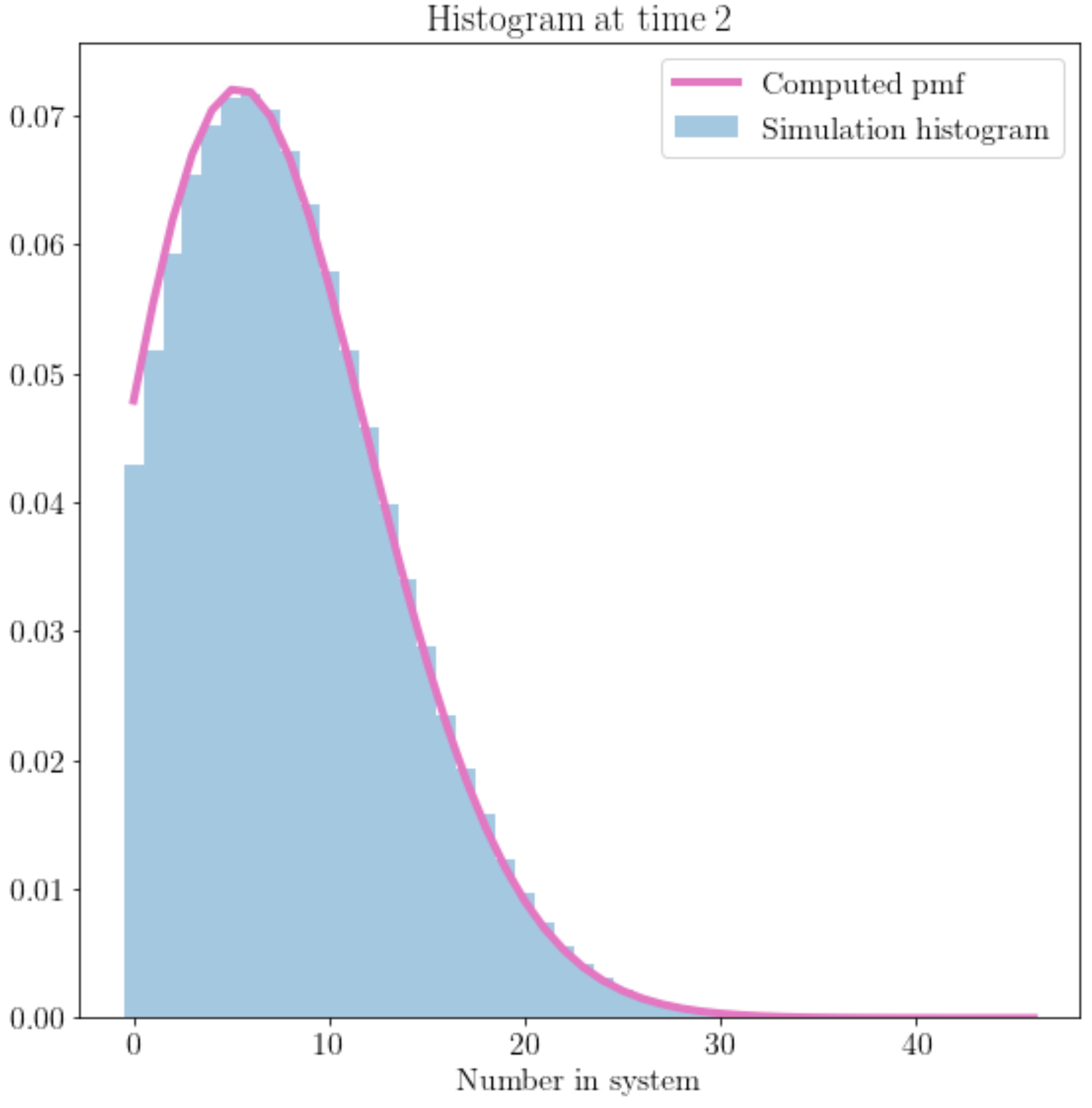
Light traffic



Heavy traffic



Overloaded



Learning and Queueing

Future Work

- Arrival and service rate values are needed
    - E.g. Join-the-Shortest-Queue (JSQ) $<<$ Join-the-Shortest-**Workload**
    - But workload needs **job size** and **service rates**
    - Can we learn them?

- New methodologies to study delay

Thanks!